International Association of Oil & Gas Producers

REPORT
2024pfh

JULY
2025

DATA SERIES

Safety performance indicators – Process safety events – 2024 data

Tier 1 PSE, fatal incident, and high potential event reports

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2024pfh

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Safety performance indicators – Process safety events – 2024 data

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Revision history

VERSION	DATE	AMENDMENTS
1.0	July 2025	First release

Contents

TIER 1 PROCESS SAFETY EVENTS (PSE)

Tier 1 PSE are predominantly lagging indicators related to Loss of Primary Containment (LOPC) referred to as a Process Safety Event (PSE). The Tier 1 KPI records events with greater consequence within the four-tier approach.

For more information see the introduction and IOGP Report 456 – Process Safety – Recommended Practice on Key Performance Indicators. IOGP has been gathering Tier 1 PSE narrative reports from its Members since 2013. These include Tier 1 PSE reported both by companies and contractors. Event reports are categorized by region, country, location (onshore/offshore), cause, activity at the time of the event, and, from 2019 onwards, point of release.

The information provided here is not detailed; often the root cause of an incident cannot be established. However, the information should assist organizations to identify likely hazards, human particular, it allows organizations to question whether their own Safety Management System would have prevented the event occurring and mitigated its consequences.

The database from 2020 onwards is available and searchable at https://data.iogp.org/ProcessSafety/Tier1PSE.

Note that a descriptive report has not been provided for every Tier 1 PSE reported.

This database is a tool for learning and should not be considered a complete record of Tier 1 PSE in the upstream oil industry or the IOGP Membership.

AFRICA ONSHORE

DATE: 04 Jan 2024
COUNTRY: Congo
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During the restart of processing chain, the unit experienced a shutdown with discharge from valve on the manifold towards the tanks. During his round, the area operator detected a loss of containment. The manifold emptied inside the containment area in the process zone.

WHAT WENT WRONG?

Bacterial corrosion and lack of supervision.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Actions needed: Bacterial Treatment Plan. Identify similar lines and implement a specific action plan. Audit the LO/L valve register.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 11 May 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Temporary
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

During routine line walk on 11/05/2024 at about 1228hrs, team reported thick smoke on the 24"/28" line right of way. The Security Team mobilized for confirmation and at about 1410hrs, the Team confirmed pipeline fire on 28" line.

WHAT WENT WRONG?

Fire/Explosion damage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Pipeline Pigging procedure review to include management of keys, isolation/ deisolation and throttling.
2. Refresh the communication strategy for pigging activity to clearly align communication between contractor, Pipeline and Asset Team.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 08 Dec 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

At about 2200 hours on 8 December 2024, Ship notified the terminal leadership of the suspension offtake at platform, due to oil sheen seen around the platform.

IMMEDIATE ACTION TAKEN:

- Export suspended.
- Further Investigation ongoing.
- Corporate ERT notified.

WHAT WENT WRONG?

A release above threshold quantity in any 1 hour period.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Include the check of marine breakaway coupling in the tanker export prestart-up checklist.
2. Provide a watchdog to monitor outage of leak and alert control room operators.
3. Work with manufacturer to indicate positive notification method when marine breakaway coupling parts.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 07 Apr 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At about 1016hrs on 07/04/2024, following leak alert from security team, Investigator's confirmation visit was carried out and confirmed crude oil leak on 24" line immediate shutdown of the 24" necessary action was advised, and leak was contained within at time of visit.

WHAT WENT WRONG?

- A release above threshold quantity in any 1 hour period

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Review Permit-to-Work with focus on use of complementary certificates (Hot Work).
2. Use of Gas Monitoring close to potential hydrocarbon source.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 01 May 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Unspecified production
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On 1st May 2024 at about 0540hrs, the sea watch alerted the security team of an attack on well. The team informed the operation team leader and a joint team comprising of the security agency, the production unit manager and operations team mobilized to scene of incident and observed oil jetting out from the well into the atmosphere/water.

WHAT WENT WRONG?

A release above threshold quantity in any 1 hour period.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. 3rd party interference to the well heads poses high risk especially tampering with surface controlled sub-surface safety valves. It has become necessary to consider more of downhole safety valves without surface interference capability.
2. Due to increase in vandalization despite our current security strategy, there is a need to improve and enhance strategy.
3. The competency and dedication of the sea watcher personnel's deployed to the guard hut are in doubt. More diligence is required in their recruitment.
4. Availability, stability and strong GSM signal is a key to the function of installed well head security equipment sending alert to the security team for security response/intervention.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

DATE: 18 Jun 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Sabotage – at about 0330hrs on Tuesday, June 18, 2024, heard a loud bang from the oil field towards the open river axis. Security team was informed about the bang and patrol team were deployed to search around. At about 0553hrs, received notification that the leak was between valve platform #1 & 2. At daylight, security, HSE er and operations went out and discovered a leak on the riser line of the 12" gas export line from VP #1 into VP #2.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 22 Nov 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Sabotage – at about 0240hrs on November 22, 2024, two consecutive loud bangs were heard from the river axis around the oil field. Joint inspection by operations, security, construction and HSE ER, identified a 3rd party breach on the 8" test line from RMP 12/18 and the 12" bulk line from RMP 12 at junction 19. The 3rd party breach on the 12" and 8" gathering lines led to spill (oil in water and on land). Estimated spill volume based on JIV report was 40bbls.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 10 Sep 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Around 2 pm, a process upset occurred at a crude processing facility. The flare, being part of the primary safety system, worked as it should with elevated flaring.

WHAT WENT WRONG?

A process upset occurred at our crude processing facility. The flare, being part of our primary safety system, worked as it should with elevated flaring

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Hardware Barrier Failures: Emergency Response Equipment and Systems

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Human Barrier Failures: Response to emergencies

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 08 May 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At approximately 0120-0134hrs the central control room initiated train 1 & 2 shutdown and shut down valve closed, offshore locations contacted to shut-in production. Emergency response team deployed fire water monitors around the tank with foam to fight fire with fire pumps confirmed running.

WHAT WENT WRONG?

Treating Plant Operator on surveillance noticed smoke with fire emanating from settling tank post several rounds of thunder and lightning strike.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

AFRICA OFFSHORE

DATE: 02 Feb 2024
COUNTRY: Angola
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

On the morning of 2nd February 2024, reprocessing from off-spec water tank was in progress, feeding the hydro cyclone system. At 5:53 AM, an operator sampled the water outlet from the PW Degasser and discovered high concentrations of oil in water (~30,000ppm). Operations responded quickly and flow was diverted at 05:57 from going overboard to back to the Off Spec Water Tank. Oil sheen was observed on port side of the FPSO.

This was a result of high oil in water content discharged to sea on main produced water Caisson. Initial calculations estimated the release quantity of oil to be greater than 10 barrels. The event was reported to leadership, notification.

WHAT WENT WRONG?

Failure to Follow Procedure / Instruction: The pre-start checks requirements to take interface level measurement & OIW sample in off-spec water tank, were not done as per reprocessing procedure.

Failure to Detect/Measure: Off-spec water tank being operated below minimum low level (1000m3) and undetected oil in water fed to produced water system.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. All the tank contents to settle for 4-24 hrs after transfer to Off-Spec Water Tank to allow the contents to separate and form two phases depending on the source, temperature, experience.
2. Measure the oil in water interface and document this well unless the tank is pumped all the way empty the oil interface level would always be increasing. Operations should look at historical interface level measurements and question how an interface could be less than previous measurement.
3. Analyse and document the safe pump stop level for reprocessing based on oil water interface and other tank sediment issues. The operating procedure says to stop the pump before 1000m3 inventory to prevent settled solids in the bottom of the tank.
4. Consider using automation to alarm and stop the pump at safe tank levels. Set a low level alarm to pre-warning MCRT to slow the transfer rate to minimum. Set a LoLo alarm to alert MCRT if the planned transfer has been exceeded and pump not stopped.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

DATE: 15 Dec 2024
COUNTRY: Congo
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During loading operation, the loading master noticed an oil sheen on the sea.

The Control Room of the Oil Terminal stopped the ongoing offloading operation to the tanker.

Leak on the 16" export hose, from an underwater pinch in one of the floating hoses (3rd section from export buoy).

WHAT WENT WRONG?

Leak on the 3rd length of the hose. Investigation ongoing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Actions needed. Integrate the action of disconnecting the hose into the oil spill scenario sheet for site. Raise awareness among the personnel involved in tanker loading. Investigation ongoing.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 23 May 2024
COUNTRY: Ghana
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During cargo oil offtake from an FPSO to an export tanker, which was moored to a CALM Buoy, the Hawser Connection Assembly (with load cell) came apart, releasing the tanker from the CALM Buoy. The export tanker drifted away from the CALM buoy, parting both Marine Breakaway Couplings (MBCs). The resulting pressure surge caused by the MBC's petals closing, initiated an executive command, actuating the shut down valve. At the time the tanker release occurred, and the MBCs parting, the pump had not stopped and the flow rate remained at the maximum of approx 33,000 bbls/hr, resulting in the unintended release of an estimated volume of 635 bbls of oil into the sea. The export tanker was disconnected and towed clear of the CALM Buoy to a safe distance (about 1nm) from all assets and support vessels immediately commenced oil dispersion by propeller agitation. There was no injury to persons.

WHAT WENT WRONG?

- Critical Factors #1: Hawser connection loadcell (pin) per design was secured at only one end (top) with a locking plate secured by 2 x M12 stainless bolts. The cylindrical sliding bearing which housed the loadcell pin was also found to be worn.

- Critical Factors #2: Hawser low-low alarm was inhibited and consequentially, the executive shut-downcommand could not activate as system logic demands.
- Critical Factors #3: Failure of Marine Breakaway Couplings (MBCs) to fully close following activation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The investigation team's key recommendations to prevent re-occurrence include:

1. Engage Original Equipment Manufacturer (OEM) to ascertain the adequacy of the primary retention system (locking plate) for the hawser connecting link.
2. Review the adequacy of the mooring hawser connecting link visual inspection and greasing checklist (Preventive Maintenance Plan) to facilitate an effective visual inspection.
3. Review of the current Mooring Master/FPSO pre-transfer checklist to ensure it adequately validates the full functionality of the Programmable Logic Control (PLC) system as per the design specifications.
4. Enhance security measures and restrict unauthorized access to the Tanker Berthing System.
5. MBCs to be returned to OEM for a detailed investigation into why they did not fully close as per the design intent.
6. Conduct a comprehensive review of the current Preventive Maintenance Plan for the MBCs and assess the need or otherwise for scheduling maintenance earlier than the 5-year conformity period recommended by the OEM.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 23 Apr 2024
COUNTRY: Ghana
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During cargo oil offtake operations from an FPSO to an Oil Tanker, an incident occurred due to the unexpected activation of one of the two Marine Breakaway Coupling (MBC) mechanisms on the offtake hose. This led to a sudden disconnection of the inner 16-inch section of the offtake hose between the FPSO and the oil tanker, and an unintended release of crude oil into the sea. An estimated quantity of 1 to 7 cubic meters was spilled. The spill was immediately contained and cleaned / prop washed by standby support vessels to safely disperse the spillage, resulting in minor impact on the environment but no permanent effect incurred.

WHAT WENT WRONG?

Based on examination of all findings and available evidence, the investigation team established that there were two critical factors identified: a pressure spike leading to activation of the MBC, and after activation the failure to close 1 out of 8 of the downstream

MBC petals. Based on the estimation of the spill quantity, it was concluded that the leak volume is >1m3 and therefore points towards a mass in the Tier 1 PSE-LOPC classification.

Due to insufficient information available to the investigation team, no specific root causes were established. The investigation team ruled out the causes resulting from any overpressure from the FPSO as these were deemed not credible. However, most likely

credible cause was Hose End Valve (HEV) malfunctioning that could result in the partial closure of downstream

HEVs. Other probable causes identified include activation of the MBC lower than the set pressure and closure of the Tanker manifold butterfly valve. A root cause failure analysis is required to determine / confirm the load at which the MBC activated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The key recommendations to prevent recurrence include defining a preventive maintenance schedule MBC changeout/ refurbishment and HEVs primary and secondary locking mechanisms as per OEM's recommendation.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 18 Feb 2024
COUNTRY: Libya
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

While the IP was performing his duties as an operator, encountered an incident. When he was monitoring the pressure of the Flash Tank, he attempted to operate an open drain valve to reduce the pressure. Unexpectedly, he heard a loud noise from the valve, which was later identified as being caused by a temporary drain line used to convoy the drained water to channels. This line failed due to corrosion and led to the connection breaking from its place. As a result, the IP suffered injuries to his face two cut wounds below the lower lip about 6 cm, the other wound inside his mouth.

WHAT WENT WRONG?

- Underestimation of risk during draining activities – As this is classified as a routine activity operator approached the job probably underestimating the risk.
- Lack of Training led to unsafe practices. All operators shall be aware of work hazards and consider these hazards seriously. The draining operations may always be executed slowly.
- Temporary modification on the open drain valve without proper management of change approach.
- Lack of Attention: since the corroded line was there since long time operators, SPVs should have been reporting criticalities (also using UA/UC forms).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- The implementation of periodical inspections and preventive action to avoid corrosion even on draining and secondary circuits.
- The provision of training and periodical awareness programs to ensure that all personnel understand the importance of following safety procedures and reporting any issues or unsafe conditions.
- Enforcing HSE procedures and, even for minor modification, get the proper approval as per company procedures (Corporate Management of Change).
- Always in draining operations, even routine ones, take proper precautions and evaluate the risk of the activity and the drained conditions (High T, High P, Gas, H2S, etc) in order to evaluate mitigation actions.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 11 Apr 2024
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

IP was cutting off corroded bolts using a grinding machine on the gas metering unit equalization line under decommissioning.

The 2" line had been fully depressurized from 105bar to 0bar, isolated by double block and bleed downstream. N2 purging and inserting of loop had been done, and gas measurement at purge outlet confirmed at 0LEL CH4 before issuing green light.

While IP was cutting the last corroded bolt, a sudden fire occurred from the flange connection. IP sustained a burn on his arms.

WHAT WENT WRONG?

Presence and release of HC from the dead leg on the 2" equalization line of stream. Cutting bolts with a grinding machine instead of cold cutting with a hacksaw. Sparks from electric grinding machine in use for the operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Leaders in key roles have an obligation to enforce company rules or seek derogations as appropriate.
- Process Isolation to comply with specific rules, based on nature of fluid, pressure and type of permit.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

ASIA / AUSTRALASIA ONSHORE

DATE: 04 Nov 2024
COUNTRY: Australia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

An uncontrolled hydrocarbon release to atmosphere occurred from a small 'Relief Valve' installed on a large pressure control valve (PV) in the inlet section of the LNG plant.

The PV is part of a set of three per LNG train and has seen extended high use service due to other parts of the inlet system not functioning properly from early operations. The release was first observed as a minor 'weep' on 2 Nov 2024 that was being monitored on the assumption that it was a similar situation to another valve from the same vendor that had a grease weep six months prior. The grease weep was previously assessed as acceptable to operate for a short period (~two weeks). The escalation potential of the PV weep, due to slightly different arrangement, was not recognised and escalation occurred on 4 Nov 2024.

WHAT WENT WRONG?

- Valve Information: Vendor manual is clear that the Relief Valve should be closed on a weep. Manual was not consulted, nor had vendor information been transferred to operating documentation.
- Escalation Potential Not Assessed: Assumed the same as similar valve.
- PV valve failure: Internal failure cause under investigation – likely contributed to by 'high use' service due to inlet throttling unavailability.
- No Preventative Maintenance: Preventative maintenance had not been conducted or planned on the valve.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Assessing risk & release escalation potential: Assessments of weep escalation potential must include all appropriate information (e.g. vendor manuals, vendor advice, etc.).
Emergency response actions should follow the PEARS approach: Removing people from harm should be the priority in emergency response actions (i.e. always consider muster to remove personnel from danger).

BARRIERS:

Hardware Barrier Failures: Process Containment
Hardware Barrier Failures: Detection Systems
Human Barrier Failures: Response to emergencies
Management System Element Barrier Failure: Risk assessment and control
Management System Element Barrier Failure: Plans and procedures
Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 Oct 2024
COUNTRY: Australia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pig launcher/receiver
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Hydrocarbon release from the pressure warning device on the pig receiver door on the gas pipeline.

WHAT WENT WRONG?

Release from pressure warning device

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 11 Aug 2024
COUNTRY: Indonesia
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Completions
MODE OF OPERATION: Completion
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Flash Fire during well service activity.

WHAT WENT WRONG?

Initial Killing is not conducted properly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Initial killing should be performed before well intervention.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

DATE: 11 Nov 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Leak on 3" flanges.

WHAT WENT WRONG?

Surge pressure occurs because of the high-pressure differential between pipe segments and sudden valve opening.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Sudden opening of valve may cause sudden increase of pressure.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 07 Dec 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Pipe rupture on discharge lean amine pump.

WHAT WENT WRONG?

Erosion on pipe elbow due to high solid particle in the amine solution.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Anomaly in the solution shall be evaluated for further impact.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 23 Jan 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

20" crude pipeline leak.

WHAT WENT WRONG?

Inspection & Maintenance program is not run properly and there is no routine pigging.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Conduct routine pigging for long pipeline and develop inline inspection program.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 13 May 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Crude heater fire.

WHAT WENT WRONG?

Inspection & maintenance program for internal tube was not conducted properly and there was short term overheating/creep rupture.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Implement deferral process if there is any postponement of inspection task.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 24 Apr 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Flash Fire during maintenance of Flare line resulting 2 people LWDC.

WHAT WENT WRONG?

Isolation Plan is not developed properly and Permit to work is not implemented properly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Use MOC if there are any changes on process plant. Implement Isolation Plan for any maintenance connected to process.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

ASIA / AUSTRALASIA OFFSHORE

DATE: 14 Nov 2024
COUNTRY: Indonesia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Smoke was observed coming from the Slop Tank on the starboard side, with pressure slightly above atmospheric levels.

The Slop Tank Starboard, water ballast tank (WBT) 3S, and cargo oil tank (COT) 4S are isolated and previously confirmed to be gas-free. There is no oil cargo in COT 3C, as it has been fully transferred to COT 2C.

No activity was taking place in the Slop Tank Starboard at the time, although hot work for re-plating was underway in the adjacent WBT 3S tank.

Upon detection, the Production Technician in the control room activated the alarm, initiated a process shutdown, and suspended all activities. Emergency response procedures were enacted, cooling the Slop Tank using the water hydrant and the firefighting system from the support vessel.

All personnel mustered at the Primary Muster Station, and no injuries were reported.

WHAT WENT WRONG?

Investigation still ongoing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation still ongoing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

NONE: Unspecified

DATE: 19 Jun 2024
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea pipeline/flowline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Operator observed huge oil sheen at the sea surface between platform A and platform B. Perform inspection at Platform A & Platform B topside for LOPC source but no finding. Bubbles bringing hydrocarbon can be seen surfacing at the south side of platform B. At 0745 hrs initiated platform control shutdown and manual ESD as platform 3 was the pipeline passing through platform A – platform B. Platform 3 topside was then depressurized in stage at platform A topside.

WHAT WENT WRONG?

The cause of the oil spill was from the sealine anchor wire entanglement, which snagged on the flange joint of the sealine during routine accommodation working barge operations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure the information used for the Anchor Pattern Analysis is the latest version.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 26 Jul 2024
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

A methanol leak at turret triggered gas alarm and ESD & blowdown activated. Gas was released from the topside methanol delivery connection to subsea.

WHAT WENT WRONG?

1. Flexible hoses were not assigned preventive maintenance and inspection in the system. Hence, their service life, inspection, and maintenance requirements were not established.
2. No Deviation and Risk Assessment (RA) to mitigate the risk of flexible hose beyond recommended service life.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- To establish Hose Management Program to address hose service life, maintenance and replacement regime, and test requirements.
- Ensure any overdue PM is properly risk assessed and mitigated as per MOC process.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 02 Aug 2024
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 2
ACTIVITY: Unspecified – other
MODE OF OPERATION: Temporary
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

Cargo pump explosion aboard the PSV which was transferring recovered "oily water" from the cargo tank into a process equipment on a platform. Seven personnel sustained injuries. Two deceased persons (DPs) and one LWDC and four First Aid cases. The "oily water" was liquid recovered from a pipeline, which contained high water cut crude oil.

WHAT WENT WRONG?

Under investigation, including PSF, PSE barrier failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Under investigation.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

NONE: Unspecified

EUROPE ONSHORE

DATE: 03 Apr 2024
COUNTRY: Austria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

Blow Out. During mounting, the borehole started to overflow.

The crew tried to secure the well – without success.

Alerting and mobilisation of the necessary emergency personnel.

After installing two preventers, the well was secured and pumped dead with 27m³ of mud.

The well was regularly checked up overnight, and clean-up work was also carried out.

A Workover Rig was replaced, the well was additionally secured with a long stroke (LS) packer and the blowout preventer (BOP) system, including the Christmas tree lower part, was completely replaced.

WHAT WENT WRONG?

Insufficient fluid in the well bore.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Exact volume calculation of well bore, sufficient fluid on location.

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 21 Sep 2024
COUNTRY: Hungary
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Line broke underground. The amount of natural gas entering the air is approx. 7000 m3. This is the first hole of the year on the section.

WHAT WENT WRONG?

Poor Design.

Quality of equipment contributed to hazardous condition.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Greater care must be taken:

- to the material quality of the pipeline transporting the aggressive product
- to prevent backflow in the well pipelines from the deployed collector

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 May 2024
COUNTRY: Italy
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

On 14-05-2024 at approximately 2:50 p.m., there was a leakage of combusted gas from a burner of the Claus train of section 300 generating the activation of the gas alarm, due to the intervention of the fixed sensors located in the vicinity, resulting in the activation of the emergency team that secured the unit and conducted environmental monitoring. While managing the emergency, a maintenance operator from the contracting firm went to the infirmary reporting that, while working in the vicinity of the burner, they had been exposed to combustion gases and was experiencing pain in his right leg. The operator, after being treated for a burn on the back of his right thigh, was taken to the hospital for further investigation. At the hospital, the diagnosis was confirmed by reporting injury with a first prognosis of 14 days.

WHAT WENT WRONG?

During the investigation, it was found that there was a non specification thread adapter between the flame detector and the connection socket to the burner, which had nevertheless ensured over time the tightness of the components to the stresses associated with the pressures involved on that equipment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- The proper use of all PPE required by the regulations of behaviour for entering the plant and in the dedicated work permit (including gloves and clothing trivalent covering all body parts) can contribute to the protection of workers, limiting the consequences of possible accidental events.
- A dedicated maintenance plan aimed at verifying visual of proper installation and verification of functionality of the flame detectors present on the burners would make it possible to capture and correct any anomalies on this instrumentation, preventing the occurrence of events similar to the one that occurred.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 02 Feb 2024
COUNTRY: Romania
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Pipeline broken between measurement skid production facility separator inside tank farm.

WHAT WENT WRONG?

Lack of risk assessment

Safe Systems of work (Permit to work / Hot work permits / permits closeout) not applied / not followed

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Enforce the application: deviation from the company standards process, MOC process for all the temporary changes made by the contractors and risk assessment process within the implemented Permit to work system. Implement a clamps management program within companies ageing facilities operated by contractors. Follow the Work Instruction Onshore Repair Technical Requirement. When deciding temporary repair with clamps keep in mind that it is a temporary solution, suitable to be used just for one year and shall follow the recommendation in the before mentioned Technical Requirement. Identification of mounted clamps in facilities and planning for replacement. Facility leaks report – correct and accurate completion, updating, distribution to all involved departments. The re-insulation of piping will be done only after the final repair training for internal staff and contractors regarding to recording of leaks and their repair.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 21 Feb 2024
COUNTRY: Romania
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Pressurised storage vessel
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Around 05:00, the R4 washing tank broke at the base and approximately 2100 cubic meters of liquid (salt water and approximate 16 tons of crude oil) was discharged inside the tanks retention dam dykes.

WHAT WENT WRONG?

- Inspection procedure not followed.
- Tank was not removed from the flow for inspection.
- Lack of risk analysis regarding the cancellation of inspections / repairs.
- Lack of risk awareness.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Updating the technical documentation and the operating manual.
- Carrying out a risk analysis regarding the integrity of the R4 tank.
- Implementation of the measures from the risk analysis, including inspection, repair or demolition of the R4 tank.
- Review the HAZOP analysis.
- Establishing a concept regarding operation, and analysing alternative measures (modernization project or decommissioning).
- Prepare a situation regarding the complete inspection of the storage tanks.
- Review the risk analysis of the tanks.
- Apply the MOC procedure for postponing planned inspections.
- Ensure the implementation of the work instruction "Inspection of vertical above-ground storage tanks".

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Commitment and accountability

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 May 2024
COUNTRY: Romania
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gas release at plant on C3+ upstream side due to a gasket failure, ESD system shutdown activated manually. Shutdown in terminal triggered controlled shutdown offshore for wells and HP Compressors, the valve was isolated and repaired. All wells went back online next day at 01:00 am.

WHAT WENT WRONG?

Inadequate hazard identification or risk assessment.

Improper use/position of tools/equipment/materials/products.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Parts/products that are not accompanied by the "Quality Certificate" and the "Warranty Certificate" will not be used.

The tightening of the bolts will be done according to the procedure and at high heights scaffolding, platforms or nacelles will be used.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE OFFSHORE

DATE: 14 Feb 2024
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A bilge pump located in the pump room on a floating production storage and offloading vessel (FPSO) leaked oil during offloading operations.

WHAT WENT WRONG?

Required leak/integrity tests specified in the inspection and maintenance program had not been carried out.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Bilge pumps to be included in the condition based monitoring program.

Consider increasing the priority and shorten the intervals between leak/integrity tests.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 19 Sep 2024
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Oil was observed on the sea during the start-up after the revision stop. Water chamber in the second-stage separator overflowed and the gauge shows 100% water level. The water outlet valve for the second-stage separator was opened. The degassing tank reached the 85% level and the valve to discharge water to the sea from the degassing tank was opened. The first sample test shows 46 mg/l. This is about an hour after the discharge to the sea began.

WHAT WENT WRONG?

Produced water released to sea was not visually checked for about 1 hour.

There was not sufficient focus on produced water in the start-up planning work. In addition, this was not discussed as a separate topic prior to the offshore start-up.

There are no automatic alerts that the oil in the process water is too high. The discharge water is based on laboratory samples three times a day.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

During start-up, it is important to visually inspect the produced water before opening to the sea.

Use and comply with SOP documentation during plant start-up or operational disruptions.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control
Management System Element Barrier Failure: Plans and procedures
Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 28 Apr 2024
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Subsea: Subsea equipment
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Methanol leakage into the sea during circulation of annulus bleed. Working on pulling of x-tree, and parallel to this, had been circulating methanol in connection with hydrates in the ring line. Challenges with hydrates in the circulation line the last few days. A leakage of methanol was observed during vertical Xmas Tree retrieval at Template B by a vessel. The spill was seen at the south side of the template. It was observed that a hot stab was out of its receptacle. The connector was reinstalled. A leak test was performed afterwards, without any sign of leakage.

WHAT WENT WRONG?

Hose connector detached at Template.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Communicate experience to well integrity and subsea responsible.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 31 Dec 2024
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Oil spillage into the sea via produced water system. Observation of discharge to the sea was reported by the outdoor operator at 20:01. Shortly thereafter, a low level was detected on the 1st stage separator water side (no low alarm received) and oil was detected from the cyclone to the 1st stage. Water outlet 1st stage was closed (20:10) and actions were taken to minimize the discharge.

WHAT WENT WRONG?

Due to slugging in the oil train, the level valve on the water side of the 1st stage separator was put in manual at 5:30 pm. The level measurement on the 1st stage separator water side decreased to 37% and flattened out. The low level alarm is set at 29% and was therefore not triggered.

Notification created in April 2024 that level gauges had to be calibrated. The job has been postponed due to capacity.

The online OIW meter has been out of operation since November.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Change the alarm threshold at the 1st stage separator to 40%, so that a low alarm will be triggered if the level should decrease.

Check the level transmitter of water 1st stage separator.

Installation of new OIW meter.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Management System Element Barrier Failure: Organization, resources and capability

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 04 Apr 2024
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea pipeline/flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A third party driving the pipeline ROW found a pool of oil and reported it to the DOC Foremen. DOC dispatched Facility MSO to verify. MSO found an active leak on the pipeline ROW. Facility MSO/Operations notified Pipeline Team for isolation and spill response.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 26 Apr 2024
COUNTRY: UK
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Hydrocarbon release from pipeline.

WHAT WENT WRONG?

Line integrity failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

No

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing.

DATE: 17 Dec 2024
COUNTRY: UK
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Incorrect valve line up leading to drains tank overflow. The loss of containment took place during an annual planned calibration of the export oil prover. System vents were opened routed to closed drains to liquid prime the prover and clear all gas, this then giving a slight rise in the atmospheric tank. The mini prover, which is tied in temporarily to the main prover tie in points were then opened to align the temporary equipment to the main system, followed by another 2 isolating valves to establish a flow loop from the main prover to the mini prover.

The contract specialist personnel had opened a valve to the closed drains system atmospheric tank, instead of the valve upstream of the drain valve (which would have provided a flow loop), this directed the diluent direct to the atmospheric tank.

WHAT WENT WRONG?

The immediate cause of the incident is lack of operator understanding of the required system line-up and the required competence to carry out related valve movements associated with the task prior to the mini prover scope commencing.

No permit was in place, the task had not been risk assessed, no vendor procedure or method statement were in place for the planned prover calibration scope. Locally it had been deemed that carrying out mini proving was an operational task, even though no S.O. document, with an agreed procedure existed, and that this task is only done on a yearly basis.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

A fully risk assessed vendor procedure or method statement attached to work permit.

Review of how movement of blinded valves is currently conducted.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 13 Aug 2024
COUNTRY: Bahrain
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil leakage from 4" flow line of well.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Isolate the pipeline to control and stop the leak, contain and clean up all the oil spill, inspect and replace the deteriorated portion of the pipeline.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 15 Aug 2024
COUNTRY: Bahrain
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil leakage on the branch of 2" flow line of well inside well manifold.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Isolate the pipeline to control and stop the leak, contain and clean up all the oil spill, inspect and replace the deteriorated portion of the pipeline.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 01 Oct 2024
COUNTRY: Bahrain
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil leakage from 4" R-Xing line of mini manifold.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Isolate the pipeline to control and stop the leak, contain and clean up all the oil spill, inspect and replace the deteriorated portion of the pipeline.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 29 Oct 2024
COUNTRY: Iraq
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Hot oil release led to fire.

WHAT WENT WRONG?

The root cause of the incident is a leak in one of the hot oil convection tubes. This leak caused oil to drop onto the ignited burners and other hot surfaces, leading to the emission of black smoke and subsequent fire. The situation escalated, requiring manual Emergency Shutdown System (ESD) activation and the shutdown and depressurization.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Always follow the MoC process to meticulously record and evaluate the impacts of any changes in hardware or operations, such as temperature, pressure, composition, and flows.
2. For any maintenance, corrective maintenance (CM), or inspection work that is deferred or postponed, it is crucial to document, implement, and monitor clear mitigation measures and controls.
3. Encourage the Emergency Response Team to plan, drill, and exercise various scenarios relevant to the plant equipment and its associated hazards or risks.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 22 Oct 2024
COUNTRY: Iraq
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Liquid carryover at cold flare.

WHAT WENT WRONG?

The main root cause of the incident is the restriction in the piping to the burn pit. This restriction was caused by the U trap acting as a flame arrestor, which resulted in increased pressure in the piping header. The elevated pressure enabled the potential for reverse flow, ultimately leading to the issues and subsequent emergency response measures taken.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The main lesson learned is the importance of effective implementation and proactive monitoring of controls and procedures to ensure safety and prevent incidents. This includes following Company guidelines, conducting proper operator monitoring, implementing previous controls, and responding to abnormal conditions promptly.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 04 Jan 2024
COUNTRY: Oman
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

12'' oil export line developed pinhole leak at road crossing. The line was isolated, and the leak was arrested. The leak discovery was on 4th of January 2024. But after investigation it was noted that it started on 23 August 2023.

WHAT WENT WRONG?

The leak occurred due to under deposit corrosion (UDC) caused by service fluid low flow and stagnation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Implement right of way procedure (ROW), which mandate daily inspection for all pipelines.
2. Implement sand removal campaign thus no sand cover pipe more than 4 weeks.
3. Strengthening pipeline integrity management activities such as (chemical injection, pigging and inspections).

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NONE: Unspecified

NONE: Unspecified

NONE: Unspecified

DATE: 10 Jan 2024
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Heat exchanger
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

At 00:30 hrs abnormal loud noise heard by operator, confirmed sound from aftercooler exchanger tube side, flammable gas detected underneath the fin fan exchanger area.

WHAT WENT WRONG?

The leak was caused by caustic (NaOH) stress-corrosion cracking. A regular cleaning agent that contains caustic was used for cleaning the external fin fan bundle.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Use no chemicals for cleaning activity on the fin fan bundles.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 27 Jan 2024
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On 27th January 2024, the facility area operator noticed heavy gas leak with sound from pipe rack nearby boiler area. The magnetic level gauge (LG) connected to header condensate drip leg in the facility completely detached and found on the ground.

WHAT WENT WRONG?

Improper design and unsupported installation of the LG during execution stage. Lack of Engineering, Procurement and Construction (EPC) vendor knowledge regarding Engineering Standard fitting requirements.

LG detached at the weld joint of 2" x ¾" branch weld joint. LG standpipe branch connection 2" x ¾" was directly welded to the pipe (Stub-on) and has much less thickness due to a bevel end which is more susceptible to weld joint failure. Stub-on/Stub-in connection is not normally used for high pressure and small branches; it is only used for cost efficiency in large diameters and low-pressure service.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Improper design and unsupported installation of magnetic level gauge during project execution stage.
- Lack of EPC Vendor knowledge on Engineering Standards fittings requirement.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 07 Jan 2024
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Other production
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A Contractor Safety Officer sustained scald burns on his right arm due to hot condensate that spilled from overhead steam line vent silencer, while he was standing on the roadside and monitoring manual excavation. The IP was admitted to the hospital for burns treatment, was discharged a week later and advised 14 days' rest after discharge. Total days lost: 21 days.

WHAT WENT WRONG?

- Incorrect and unauthorized operation of control valve 51-PV-010 instead of 51-PV-010A via mimic panel.
- Absence of controls to prevent inadvertent operation of critical valves.
- Maintenance lapses observed in control valve 51-PV-010, resulting in lack of overall risk understanding and appropriate mitigation requirements.
- Inadequate consideration with regards to safe design and location of vent line for hot condensate release.
- Failure to consider the risk of hot condensate release and lack of physical barrier to restrict personnel near vent/silencer.
- Ineffective supervision of contractor activities by Projects and Refinery teams.
- Adequacy of drain line for condensate in vent line after silencer installation.
- Permit-to-work system complacency, together with local custom and practice leading to insufficient oversight of contractor activities.
- Deficiencies in access control protocols for industrial control systems and restricted areas.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Formalize and control Engineering Control Room (ECR) access. Direct supervision of all activities in ECR/critical sites by company staff.

- Evaluate silencer vent design adequacy for all operational scenarios.
- Ensure compliance with approved method statement by contractor and requisite supervision during all project stages.
- Assess drain nozzle sizing on existing silencer.
- Include condensate release risk through vent in refinery risk register and implement adequate mitigation measures from risk assessments.
- Follow Management of Change process for permanent drainage facility installation.
- Reinforce strict compliance to Life Saving Rule relating to Permit to Work and site procedures.
- Review assurance processes for PTW and procedural compliance.
- Repair passing valves promptly.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 17 Apr 2024
COUNTRY: UAE
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On 17th April 2024, a pressurized 20" sales gas pipeline was exposed to sudden water flooding due to adverse weather conditions that caused section of the pipeline (approx. 140 m length) to bend, and subsequently ruptured. A low-pressure alarm (below detectable range) activated, and a field operator was dispatched to isolate the line from the nearest isolation point which was non-accessible due to water accumulation. Nevertheless, pipeline succeeded to isolate 20" pipeline at the pipeline launcher and station that controlled the released gas.

WHAT WENT WRONG?

Operating at an abnormal situation due to flooding/climate change with precipitation levels of 254 mm/Hrs compared to the existing design standard which specifies 48 mm/Hrs as design basis.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

MIDDLE EAST OFFSHORE

DATE: 04 Jan 2024
COUNTRY: Israel
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Loss of Containment on recycle pumps resulting in spill of Rich MEG to deck at 270F, above flashpoint (140F).

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 28 Jun 2024
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Reactor
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Following a corrective maintenance activity carried out during the day, the Programmable Logic Controller (PLC) of the Electro-Chlorination Unit was reset. After the reset, a violent explosion occurred in the Electro-Chlorination Unit, causing a blast and multiple injuries to the Injured Person (IP) who was in the immediate vicinity.

WHAT WENT WRONG?

There was one or several sources of leakage in the enclosure that caused a release of hydrogen inside.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Management of Change System Needs Improvement.
- Hazard analysis Needs Improvement.

- Design Not to Specifications.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 09 Mar 2024
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Filter
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A gas leak occurred at an offshore platform on the Riser Platform Cellar deck, prompting mustering procedures to be executed. All toxic and flammable gas detectors at Train-1, Module-1 activated. Another set of detectors activated ESD. Deluges released. Rapid Response Team was deployed and reported gas cloud formation near Train-1 condensate filter area, and a significant leak from condensate filter top cover securing clamp. No personnel injuries.

WHAT WENT WRONG?

1. A lack of in-depth analysis and insufficient end user technical involvement in the decision to change procurement of the material from the OEM to a local supplier.
2. Improper design of the O-Ring since the O-Ring size versus groove size do not comply to several applicable standards.
3. Lack of maintenance strategy to determine the wear and tear failure mode on top cover mechanism.
4. Train-1 Condensate Filter drained, top cover opened, and 'O' ring found damaged due to possible material degradation.
5. Threaded tie-rod for top head opening mechanism found distorted and inoperable.
6. Quick clamp mechanism was in inoperable to ease operations of filter housing top cover and securing O-Ring.
7. Specific method statement for the opening of top cover and replacement of condensate filter elements was not available.
8. Observed heavy flat head was opened with the assistance of rigging gear/chain block.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Procurement team to communicate with end user in case of changing the procurement sources from OEM to Non-OEM material. Accordingly, technical review should take place by the end user with Offshore Engineering Discipline team.
2. Open the communication with OEM about:
   a) The standard being used to determine the O-Ring size in correlation to groove size.
   b) Capability of OEM to supply the O-Ring material.
   c) To verify that the O-Ring material supposedly capable to handle H_2S content.
3. Overhaul and replace the top cover component mechanism to ensure the alignment between top cover and filter head flange.
4. Review the maintenance strategy of top cover mechanism to identify wear and tear through FMEA or RCM analysis.
5. Ensure and always follow O&M manual instruction on replacing the O-Ring.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

NORTH AMERICA ONSHORE

DATE: 28 Jul 2024
COUNTRY: Canada
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Drilling
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A fire occurred at field, on rig in the shale shaker and degasser area.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 28 Jan 2024
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Plant tripped on low pressure flare knockout. Operations found hydrocarbons at base of stack. Further investigations determined two water tanks carried over as well. Total calculated volume carried over is estimated at 5.74m3.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 18 Jun 2024
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Steam circulation was initiated on a new SAGD pad to establish communication between wells during the start-up phase. Approximately 50 days of steam circulation were required. Erosion was identified as a risk, and an erosion monitoring program was set to begin three weeks after the first steam due to a previous risk assessment aimed at minimizing worker exposure to rapid thermal expansion (wellhead jumping) during this phase. Approximately five days into the process, the injector well began returning fluids, and about two weeks later, severe piping erosion led to a failure and release of emulsion. The failure was discovered by an operator during routine rounds approximately 10 hours after the release began.

WHAT WENT WRONG?

Delayed implementation of the erosion monitoring program, combined with a quicker than expected erosion rate which allowed severe piping erosion to occur undetected. The late discovery of the failure further exacerbated the situation, leading to a significant release of produced fluids.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Existing programs were unable to detect the increased erosion risk following adjustments to the monitoring program. The monitoring programs were adjusted several years prior to the incident to protect personnel from the hazard of wellhead jumping during startup phases. Improvements to wellhead design, including metallurgy and process control (e.g., control valves), can improve erosion resistance and reduce the likelihood of failures.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 21 Jun 2024
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Operator received a radio call from a Truck Driver that there was a lime spill at the lime silo and the hydrated lime truck driver needed assistance.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

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DATE: 23 Feb 2024
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During the startup of 4P3212 after maintenance, the seal failed releasing diluent into the P4 DRU building. The pump was immediately shutdown.

WHAT WENT WRONG?

The pump seal failed upon startup after maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 02 Dec 2024
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Drilling
MODE OF OPERATION: Drilling
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

A fire was observed near the blender area. Emergency shutdown devices (ESDs) were activated on all frac equipment and adjacent wells were shut in. Crews were mustered off location, and all personnel were accounted for without injury.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 14 Jun 2024
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A reportable release was discovered on 14 June 2024 at the wellhead. An agricultural tractor struck the wellhead resulting in an uncontrolled event. An unknown volume of fluid was released. Clean-up and investigation operations will commence once the area is safe. This location lies within a severe winter range area high priority habitat.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 26 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Facility lost power and vapour recovery units went down. Lightning struck one of the condensate tanks. The thief hatches blew off and fire was coming from stack vent and thief hatches on 2 tanks.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Ignition Control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 15 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Oil release out of water tanks C and D due to emulsion/foam escaping through E hatches when bringing pad 42 wells online released 9.97 bbl of oil and 1.429 bbl of produced water to land.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Communicate between different functions effectively

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 13 Aug 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Unspecified production
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

While flowing down the in-well in preparation for P&A, the frac tank was overfilled. This resulted in a spill of 13.6 bbls of crude oil onto the well's pad.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Detection Systems

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 08 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Contracted mechanics were completing maintenance on compressor when a flash fire occurred causing burns to body of mechanics. The IP was transported to the emergency room by fellow mechanic and then air lifted to burn centre. IP was released the following day.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Post incident, assign a company representative to remain with vendor/contractor during their teardown process to observe any outcomes associated with their investigation.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 09 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

A 4" gas lift flowline for the well failed in the middle of a 90 degree fitting on the flowline inside below grade well head cellar, causing 29 MSCF (865.9 kg) natural gas to release to atmosphere.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Erosion or corrosion, combined with a lack of a consistent inspection/monitoring program, created a latent condition that eventually led to the pin hole leak failure of the 90 degree fitting.

Sampling coupon detected corrosion requiring action; however, data was not acted upon or lost when well was converted from an ESP to Gas lift.

Gas lift conversion design did not account for the sampling coupon.

Control room operator is tasked with monitoring multiple wells (700+) simultaneously, which makes monitoring trend data difficult.

Well is designed to be shut-in or brought online remotely. Lack of remote indicators or delay in physical operator round checks increase the likelihood of losses of containment occurring for a longer duration.

Leak was detected and isolated early only due to a separate and unrelated activity.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 29 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A hole developed on the weld of an oil gathering pipeline, resulting in 248.5 bbls of crude oil to release to land.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment
Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.
Human Barrier Failures: Surveillance, operator rounds and routine inspection
Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 14 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A 2'' load line was left open after 526 bbls of crude oil was stolen from a tank battery. This caused a spill of 102 bbls of crude oil & 1 bbl of produced water onto facility's pad and to adjacent habitat.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 07 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Operations alert at the early life compressor. The facility ESD by the fixed air monitoring. Operations shut down the facility remotely. Upon investigation, a 3/4" stainless steel tube separated from the third stage scrubber discharge. Total CO_2 1,18 lbs. Total natural gas 250 lbs., along with half a gal of condensate.

WHAT WENT WRONG?

Tubing failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 25 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The well experienced a failure of the casing flowline which allowed for 0.1 barrels of oil, 0.4 barrels of produced water, and 17,231 total lbs of gas (15,220 lbs of CO_2) to be released. The failure consisting of a split/crack on a ~20" flanged spool occurred for 24 minutes before it was safety isolated by responding personnel. This is an unmanned remote work location with no personnel present during the event.

WHAT WENT WRONG?

Casing flowline failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 25 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Flowline leak near the Manifold 3 receiver resulting in loss of containment of sour gas / inlet feed gas. The flowline was not producing, but contained 65% CO_2 / 21% CH4 / 8% N2 / 5% H2S / trace C2H6, He, and water at 1050 psig / 60 deg F (est.) at time of release. The 8" flowline was buried 4" underground. The leak occurred approximately 4.6 miles from the gas dehydration system.

WHAT WENT WRONG?

Control valve failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 08 Sep 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The Control Centre identified a pressure drop on the Early Life Compressor and H2S and LEL levels began to rise. The Centre initiated a facility shut down via ESD. The ½" stainless steel tubing had disconnected from the 1st stage suction scrubber. 473 pounds of CO_2 and one quarter (1/4) barrel of condensate was released inside the building.

WHAT WENT WRONG?

Tubing failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 31 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Flange leak was identified on the Injection Flowline 3-14 between two valves on a vent to the process flare, resulting in process loss of containment outdoors. A fixed gas detector located ~250 feet away detected 80ppm. The line was shutdown to isolate the leak.

WHAT WENT WRONG?

Leak between two valves.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 30 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Pressure transmitter that controls the Train 1 inlet pressure valve (PV) to flare provided a false reading that caused the pressure valve to open sending ~280 MCFD of inlet process gas (5% H_2S) to flare.

WHAT WENT WRONG?

Release of gas to flare.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

DATE: 27 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The construction lead on-site notified operations of an audible gas leak inside of the early life inlet compressor building. The operator arrived and ESDed. He called for backup and safely completed isolation from outside. After the building was safe, the operators identified a crack in 3/4" piping between the compressor bridle and the scrubber.

WHAT WENT WRONG?

Piping failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 16 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Filter
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Train 1 Inlet Filter Coalescer resulting in process loss of containment outdoors.

WHAT WENT WRONG?

O-ring failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 10 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Approx. 7,389 lbs of CO_2 was released from 3" carbon steel discharge line on the line heater going to the test separator inlet (initial release and blowdown lasting ~20 min). Operator received a Low pressure alarm and remotely activated the SSV on the WHU 1311 (in test). No known off-site impacts or immediate hazards in area identified.

WHAT WENT WRONG?

Discharge line failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 04 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Completions
MODE OF OPERATION: Completion
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

While drilling out frac plugs, the flowback manifold was aligned to flow through the mud-gas separator to flare. Gas was flared for approximately 5 minutes before a flash fire occurred near the mud-gas separator. Three crew members sustained burns as a result of the fire.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 09 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

A rotating equipment tech was responding to a low discharge pressure alarm when they determined the third stage fuel gas conditioning line had cracked at the threaded connection, resulting in a release.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 09 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Pipeline manifold pressured up and came apart releasing gas and a mist of condensate with methanol.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 03 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Night operator was dispatched to compressor shutdown. Upon arrival he found a very small incipient fire on a plastic wiring harness on the engine and on the air intake rubber boot, and signs that a larger fire had occurred but self-extinguished.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 15 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator arrived at facility at 1:30PM to find the 2" gas lift line directly upstream of the control valve had parted completely.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 11 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

The operator visited site to troubleshoot Train 1 H2S analyzer that was showing a relatively flat trend in the software platform. The operator noticed the pressure regulator feeding the analyzer had zero pressure and the allen screw/bolt was found to be loose. Operator proceeded to tighten regulator bolt to enable pressure flow to regulator and gas started releasing from weephole in regulator. The allen screw/bolt was loosened again by the operator and gas release stopped.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 14 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Crew was working to drain vapour recovery lines when they u-tubed and drained out downstream line. Spill dimensions 90'x6'x1/4" Total volume 11 bbl. recovered 9 bbl. leaving 2 bbl. soaked in, all oil.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 29 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Pulled up to location and noticed an oil release in containment coming from the circulating pump and noticed that the ball valve was left opened.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 29 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Tank had a reported 331 bbls of oil. Trucking pulled remaining oil down to load line from tank 2 and pushed into tank 3. By then, gauge of tank 2 was at 5'5". While picking up oil off the liner it was noticed that there was fluid underneath, and a patch of oil had popped up separate from the spill further down the liner. Oil picked up by Vac truck deposited into water tank 1. Estimated 150 bbls spilled from the tank. 171bbls was deposited into tank 3 and approximately 74.52bbls picked up and deposited into produced water tank 1.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 13 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Night operator discovered pump containment spilling over. This resulted in approximately 59.86bbls being released inside lined containment and 1.82bbls being released on pad.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 09 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Pipeline developed a leak. During rounds, fluids were found in the pipeline right of way (ROW). Proper notifications made and estimated 230 bbl. of oil released on the ROW.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Sep 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Operators were having issues with the service provider running their pumps due to a generator being down on their side. Throughout the night before morning spill, operators contacted the service provider, who sent out a mechanic to work on the generator. Around 11:15 the mechanic updated operators that the generator needed parts. At 12:00 AM compressors were shut in. The operator made rounds after shutting in wells and compressors and found fluid coming out the tops of the oil tanks. The tank levels were 27.8" and Hihi shutdown is 28.5'. Estimated release volume-30.0 BBLS of produced oil. Estimated duration of release-15-20 minutes. All fluids remained inside lined secondary containment and was recovered.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 09 Sep 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Fire tube gasket on the heater failed releasing 58.5 bbls of oil and 10.3 bbls of water.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 25 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

1" check valve on chemical roll line parted on top causing spill.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 16 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Oil tanks overflowed into containment, and some over sprayed onto the ground.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 01 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator found a 1" roll line broken off from oil tank at the Battery. He was able to get the old threads out of the tank and replaced the nipple and valve to isolate the leak on the tank. The oil was sitting on rainwater inside secondary containment from the day before. We picked up the standing fluid (oil and rainwater) and let it sit until it separated. We picked up approx. 30 barrels of oil with vac truck. Operator checked the battery at 8:30am and found the leak at 11:30am.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

The operator received an alert from the Facility that there was a problem. He immediately shut the water inlet remotely to stop the flow into the facility. He then went to location and discovered a total of 86.3 barrels of condensate spilled inside the unlined gravel berm with 70 barrels recovered via vac truck. Investigation revealed that a programming error caused the high alarm to go to the "unknown" folder in the monitoring system, so the monitoring system and operator did not receive the high alarm. The high-high shut down failed because the pressure sensor was set for the specific gravity of water, so when the tank filled with oil it read the level as lower than the shutdown set level.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 17 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The operator was on location when he witnessed an oil leak develop inside the facility. Upon isolation and further investigation, the operator discovered that the 2" check valve top came off at the edge of the blow case skid.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 15 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A 3" ball valve connected to the drain line of the heater treater separated, resulting in approx. 16 barrels spill of oil/water mix (14 barrels on pad, 2 barrels off pad). With the loss of the pressure on the heater the battery ESD shut down both wells. Spill is both within the containment, on pad, and off pad.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 26 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

A contractor managed general maintenance crew was dispatched to clear debris from a 2" compressor start gas line with a pressure washer. Initial work authorization and energy isolation was approved by personnel prior to starting work. Upstream and downstream valves were closed and locked, but the crew did not fully de-energize the system. The IP opened a 2" ball valve near the open end of pipe. At this time high pressure gas was released, the unsecured pipe moved, and gas impacted the IP on the right side of his face. The IP was transported to a local hospital where he was treated for debris in his eye and was later released.

WHAT WENT WRONG?

Contractor deviated from work scope and intentionally trapped gas in the line resulting in a gas release and subsequent injury. Pressure in the line was underestimated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Work scope provided to require additional job-specific instrumentations.
2. Standardize work authorization process to prevent errors on contractor's Lock out tag out LOTO documents.
3. Update control of work authorization process to confirm proper valve alignment for LOTO.
4. Standardize competency management program for assessing contractors.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 06 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

A contract compressor mechanic (CCM) arrived at the compressor station and immediately discovered a gas release. After moving crosswind and upwind of the release, the compressor technician identified the leak location. The CCM shut down the compressor and isolated the leak location before notifying personnel.

WHAT WENT WRONG?

After reaching the back of the compressor, the CCM noticed the broken connection between the 2-in start gas line and the 8-in suction line on the compressor. The suction line on a compressor was known to be a high source of vibration off the skid.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Lead process safety advisor to revise the facility construction manual to include guidance on requirements for welded connections between compressor skids and piping.
2. Asset integration process safety team (AIPS) to update the integrity management policy to include guidance on vibration testing and inspection requirements for large compressors.
3. Management systems specialist to revise the routine wellsite checklist to include excessive/severe vibration of piping.
4. Project and facility managers to verify all facilities with multiple compressors meet vibration testing requirements for off skid piping from integrity management policy.
5. Facility engineer, in collaboration with category manager, to ensure compressor contracts are updated to include timeframes for corrective actions from vibration studies to be completed.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 05 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

A production line to the battery tanks was prepared for maintenance to replace several valves. The valve used to vent/bleed oil from the line was left open while the maintenance work was completed. The release occurred when the valve used to vent/bleed down the production line was not closed prior to putting the line back in service for leak testing.

WHAT WENT WRONG?

Vent valve was not closed prior to putting the line back in service.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Training on Start Work/End Work check must emphasize that these checks are individual task checks, not daily start/end task checks.

Sanction to test (STT) operations must be accomplished with the same emphasis as initial LOTO placement and final LOTO removal.

Sanction to test LOTO manipulations need to be recorded just like initial LOTO and LOTO removal.

BARRIERS:

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 26 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

A thermowell on the 1st stage discharged piping of an ethane compressor cracked and leaked.

WHAT WENT WRONG?

Engineering review of the thermowell could not definitely identify the cause of the cracking but most likely cause was high cycle fatigue.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The thermowells on all the ethane compressors were changed to a shorter length to reduce the likelihood of future high cycle fatigue.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 29 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Unspecified production
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

A production facility was down because the heater/treater was out of operation. The storage tanks caught fire when the flare backflashed through the tanks' vent header.

WHAT WENT WRONG?

Flare ignitor was left in service when the production facility was isolated in preparation for an extended downtime. The flare ignitor was the ignition source of the fire.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Procedure for long term tank isolation needed additional clarification on flare ignitor status.
- Operators did not fully understand the consequences of leaving the flare ignitor in service when the battery tanks could have oxygen in the head space from tank "in breathing" because there was no blanket gas on the tanks.
- Installation design reviews are required to add safeguards to prevent flare backflash.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 05 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Unspecified production
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

On 5 February 2024 crews at the CGF facility were performing maintenance on an oil separator vessel. A crew of four roustabout workers, from 1888, was removing a 6" drain spool piece from the bottom of a separator vessel. After removing a majority of the bolts from the top and bottom flanges the crew shifted the 90 degree spool piece. Upon shift of the equipment, a release of oil, condensate, and water mixture exposed the workers. No pressure was on the vessel. Release of fluid was gravity fed out of the top flange. Workers stopped and evacuated the area. After removing saturated clothing, the workers cleaned themselves and reported exposure to HSE. No injuries reported. No damaged equipment reported.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 19 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A piping failure occurred at a tank battery location. 203 bbl of oil and 57 bbl of produced water was released inside secondary containment. Equipment was shut-in and clean-up efforts are ongoing.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

A 1/2" bleeder valve on the oil dump of the train 2 heated production separator (HPS) was open, resulting in 124 bbls of oil to release to land.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Digital solution not available (bull plug and valve misalignment).
2. Provided feedback to standard facilities design team on the valve design/size and location of valve (if intended for sample point and how it can easily get plugged).
3. Formal communication standard not deployed at location.
4. Task familiarity is thought to be picked up during onboarding/on the job training/training.
5. Operator works his shift under "manage by exception".
6. Work schedule over at 5:00 pm and night rider schedule starts at 6:30 pm.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 22 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

An equipment failure on a vessel at the off site tank pad occurred causing a release of 170bbl of oil/water mix. 102bbl oil and 68bbl of produced water. The release was contained on location with-in the steel walled area. Equipment was shut-in and clean-up is underway with currently 127bbl of fluid recovered.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 28 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Not applicable.
INCIDENT DESCRIPTION:

While pulling down off-spec tank with a vacuum truck for asset integrity maintenance, the field supervisor noticed the tank leaking from the bottom. Leak was not present when they initiated the tank pull down.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Increased potential for similar incidents to occur due to same design associated with multiple tanks at all company locations.
2. Design prevents additional and more specific internal inspections of the tank's piping.
3. Tank and piping meant for "clean" service. Uncoated piping not designed for off-spec product.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 18 May 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

An agricultural vehicle (sprayer) struck the wellhead on 18 May 2024 at approximately 7:38 PM (time of 911 call).

Based on a braden head test conducted on 7 March 2024 casing pressure was 1466 psig and tubing pressure was 1465 psig. This pressure, 1466 psig is the pressure assumed at the time that the wellhead strike occurred. The orifice size is 1.995 inches and gas vented from that orifice for 18 hours. The well had 46 psi on it when the well was capped at 1:35 PM on 19 May 2024.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

NONE: Unspecified

DATE: 15 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

At approximately 1600 on 1/3/2024 the pig was received at the site. At that time, the inlet PSV lifted prematurely and did not re-seat, sending the feed to the flare 1. Resultant was a TEG/water rainout that misted up to ~1300ft in the SW direction from the Flare. H2S Parts of ~100ppm were seen at numerous locations on fixed monitors. Ambient conditions were 32F and low wind from the NNE. Personnel mustered and all persons were accounted for.

The release resulted in a Tier 1 PSE (above the 55lb Tier 1 threshold for H2S). PSV was set at 1155PSIG. Temperature was 97F.

WHAT WENT WRONG?

PRV Release.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 13 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Contractor discovered a sample valve on bottom of a 6" transfer to the charge pump was open, releasing fluid. Operator stopped the release by closing the valve.

WHAT WENT WRONG?

Contractor discovered he had inadvertently bumped the valve open during his previous visit which caused the release.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Focus on increased awareness while on site.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 14 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Operator pulled onto location finding a busted load line due to freezing. The oil tank valve also failed allowing the contents of produced oil tank to release.

WHAT WENT WRONG?

Freezing water broke multiple valves.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 14 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

A pilot operated PSV on the bulk separator and the heater treater froze and malfunctioned causing fluids and gas to release.

WHAT WENT WRONG?

PSV pilot line froze in weather event.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: 16 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Operator pulled onto location finding oil tank load line valve had failed due to freezing.

WHAT WENT WRONG?

Valve failure due to freezing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 15 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

A pilot operated PSV on the bulk separator froze and malfunctioned causing a release of oil and gas.

WHAT WENT WRONG?

Freezing in pilot valve.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: 18 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Operator found the tank had released liquid to the impermeable containment due to a broken hammer union on the dump line caused by freezing weather.

WHAT WENT WRONG?

Freezing Weather

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 17 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Operator discovered a check valve on the main water line had failed due to the freeze causing fluids to back feed into the separator filling it up and out the Pilot operated PRV.

WHAT WENT WRONG?

Water line froze.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: 20 Jan 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Operator arrived on site and noticed gas venting from a broken nipple on the glycol pump.

WHAT WENT WRONG?

Nipple on dehydration pump failed due to vibration and freezing temperatures.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 23 Feb 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

1/2" bleeder valve found open below the level transmitter on the oil tank.

WHAT WENT WRONG?

Failure to follow written procedure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 15 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Operator found that the facility was shut down and noticed oil releasing from oil tank. Operator closed the manual block valve positioned upstream of the control valve and shut down valve to prevent flow coming into the facility.

WHAT WENT WRONG?

Automated high, high call was not received. High, high critical alarms just sent a text message instead of a call out.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

DATE: 31 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Upset
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Technician discovered overfilling tanks. The cause of the spill was a result of power loss leading to alarm and communication failure.

WHAT WENT WRONG?

Facility logic and critical fluid level instrumentation not functioning properly or reading accurately

Facility ESD Valve was in manual mode and not automatic, manual mode was a simple "switch" which is easily missed

Manufacturer updated firmware on control panel which experienced numerous faults within the update

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Implement a software review and testing for all software updates prior to deployment.

Confirm software updates have been correctly implemented at other sites.

Engineering review and Install level override functions in storage tanks to prevent overflow incidents and implement regular maintenance and testing procedures for these systems.

Add locking mechanism to ESD valves so that manual mode requires additional operator intervention to switch into manual.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 30 Mar 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Operator found the water dump valve washed out on the bulk vessel sending gas to the skim tanks and causing gas to escape out of the tank vent.

WHAT WENT WRONG?

Control system failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 12 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator discovered an NGL leak occurring in the product storage area due to a compromised valve.

WHAT WENT WRONG?

After further investigation it was determined the leak was occurring from the body of the valve.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 01 May 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Technician discovered a release at station due to tank overflow.

WHAT WENT WRONG?

Facility inlet shutdown valve did not function as designed due to undetermined internal failure.

Lack of proper UPS systems to allow for critical equipment to operate when power loss is received at the facilities.

Operations readiness review of facility prior to startup could have identified equipment and design deficiencies prior to operations assuming custody.

Operations delayed response to facility alarms worsened the incident magnitude.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Define clear responsibilities and expectations for on-call personnel when receiving alarm notifications.

Engage with engineering teams to reassess the system design and identify any potential flaws. Address design issues promptly to prevent similar failures in the future.

Conduct a thorough review of the emergency shutdown valve's configuration. Implement necessary adjustments or updates to ensure proper operation.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 30 May 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Technician discovered a release at the tank battery.

WHAT WENT WRONG?

The cause of the spill was a result of power loss leading to alarm and communication failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 12 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator reported a leak from a 400-bbl. storage tank into containment.

WHAT WENT WRONG?

Integrity of tank failure due to internal corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 02 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

After bringing a well on-line, the operator noticed a leak coming from a 4'' pipeline.

WHAT WENT WRONG?

Leak due to internal corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 09 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

While bringing wells back online after a power outage, operator found a pin hole in the 4'' low pressure flowline.

WHAT WENT WRONG?

Pin hole due to internal corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 29 Jul 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Broken nipple on the dehy piping caused a gas release.

WHAT WENT WRONG?

Possible fatigue related failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Aug 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Upset
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Site experienced an operational upset resulting in ~70 bbls of oil exiting the low-pressure flare and igniting. Repairs to flare were required.

WHAT WENT WRONG?

Following unconfigured device addition to active network, configuration issue caused software fault which prevented wellpad shutdown when needed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Verify Comm Fail interlocks are enabled at all locations with similar network setups.

Integrate commissioning procedure for adding new devices to an active network

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 19 Aug 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A spill occurred when the 4" v-ball washed out and ran skim tank over.

WHAT WENT WRONG?

Valve malfunction due to lack of preventative maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 22 Aug 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

While pressurizing up after unrelated maintenance had been performed, an operator discovered the 6" inlet valve leaking where the valve body is bolted together. Inlet line was immediately isolated and blown down.

WHAT WENT WRONG?

Valve had a faulty o-ring.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 21 Aug 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Maintenance crew was emptying an isolated bulk separator with a vac truck and hooked up to a valve on the bottom of the vessel that had an unknown internal stand-pipe (error prone situation). Once they finished pulling fluid through this connection, the crew removed the man-way as they believed the vessel to be empty and oil was released.

WHAT WENT WRONG?

There were multiple valves on the vessel. No valves were labelled and the crew was unaware to which valve the skim pipe was attached.
This vessel had a different manufacturer from most of the other separators. Due to the difference in manufacturer, the skim pipe was on a different valve than usual.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity
Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 10 Sep 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

A leak was discovered by a third party and called in. Personnel were dispatched to locate and identify the leak. The leak was identified as in a compressor discharge line.

WHAT WENT WRONG?

Leak had developed in a scab that had previously been welded on the line. Internal corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 15 Oct 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

1/4" nipple on high discharge knockout broke releasing natural gas into the knockout building, triggering the LEL alarm and shutting the site down.

WHAT WENT WRONG?

Vibration/harmonics broke nipple.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 06 Nov 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator discovered leak on emulsion line running from satellite facility to tank battery.

WHAT WENT WRONG?

Leak was due to internal corrosion around a 2" tapped Victaulic blind causing produced hydrocarbons to be released.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity
Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 09 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

While dumping the sand knock out (SKO) manually for sand evacuation, frac tank overflowed fluids onto pad. No injuries.

WHAT WENT WRONG?

Each producing well on the pad has 2 sand knockouts per well. Contractor closed all valves on one well but failed to close all manual dump valves on the other wells.
Contractor thought that opening all the SKOs at once was easier/faster/better than performing the task one at a time. Contractors was trained to perform one SKO at a time and all other operators stated they perform the task one at a time.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Contract company has had discussions with all crews regarding SKO process, allowing workers to lead the discussion detailing job steps then ended discussion with company's operational goals.
Contract company will have a meeting at least monthly in which discussions over company promoted or situational checklist guide with issues ranging from unfamiliarity with tasks to re-iterating daily habits and expectations.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.
Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 13 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A mechanic was walking through the compression building when they noticed the white warning light on the unit LEL monitor alarming on high LEL which triggered the individual to muster.

WHAT WENT WRONG?

The leak occurred from a failed diaphragm.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 22 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Operator discovered a damaged flowline, then shut in the well. During follow up, determined a 3rd party equipment operator had the rippers of the grader in the down positions when they encountered an obstruction on pad causing them to detour off pad and made contact with a flowline.

WHAT WENT WRONG?

3rd party vehicle impact with flowline.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

DATE: 07 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Meter
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The sealing bar O-ring on the flow meter failed and caused a gas leak.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 29 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

Upon Start-up of the facility after shutdown it was discovered the drain valves located in the skirt of the Interstage Suction Scrubber were leaking by internally to the closed drains. The valve crew was called to service valves. While servicing the single plug root valve the grease fitting which had a previously installed piggyback fitting started leaking.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 28 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During start-up operations of the gas injection system post turnaround a fire and gas alarm came in. Upon investigation by the operator it was determined the leak was coming from the body flange of remote operated valve.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 10 Jun 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Flange downstream of choke was found to be leaking process gas (MI).

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 08 Dec 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Operator responded to facility shut in on LSHH for heater treater on the oil side. The alarm had cleared when the operator arrived on location, and they started bringing the facility back online. The operator started blowing down compressors for start-up. After opening the blow down to flare valve, the operator walked to the control panel and heard a snap sound. He turned and noticed a vapor cloud coming from the blow down line moving toward their truck.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

NORTH AMERICA OFFSHORE

DATE: 04 Apr 2024
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea equipment
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Ongoing activity by Project was to replace the fluid sampling point (FSP) blind stab on the flow control module on well by use of Remotely Operated Vehicle (ROV). The 4 barrier valves on the 2 ea. 1" pipes to and from the stab housing on the flow control module were visually verified by the ROV operator to be in closed position before initiating removal of the FSP blind stab. When removing the blind stab, a subsea gas leak was observed. Well was shut in after 10 min, and the leak rate was reduced. The header valve downstream on the manifold was closed after 42min, and the gas leak stopped. No gas condensate detected or observed on the sea surface.

WHAT WENT WRONG?

Not Available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 30 Mar 2024
COUNTRY: Azerbaijan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Abandonment
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

In an abandoned well uncontrolled gas flow occurred which resulted in a blow out and jet fire. Incident investigation found out illegal interference.

WHAT WENT WRONG?

Lack of site control due to long distance to the well site from nearest company facility.

Inadequate process safety management.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Inspection of the decommissioned and abandoned wells to identify and assess the risks.

Preparation of a long-term action plan to ensure the sustainable protection of the wells.

In order to protect the territory, facilities and equipment from illegal interference, the area around the abandoned wells should be fenced off and provided with relevant safety signs, to prevent the entry of unauthorized persons and any type of plant equipment without a special permit.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 14 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Upset
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A gas blow-out occurred in the well when a contractor mobile crane hit and damaged the Christmas tree of the well while manoeuvring on the site.

WHAT WENT WRONG?

Inadequate site control.

Inadequate contractor management procedure.

Inadequate control of work and risk assessment procedure.

Lack of competency.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Contractor management procedure to be enforced.

Safe driving on site rules to be enforced.

Awareness sessions to be conducted with all involved crew.

BARRIERS:

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

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DATE: 21 May 2024
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A release of crude and gas occurred at meter station from a weld joint on a 20-inch gathering line. The leak was identified by the activation of a fixed h2s detector, detecting 51ppm. The line was installed in 2020 but the section where the leak occurred had not been placed in service. Early data gathering indicates that the valve isolating the affected section was not fully closed.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 05 Sep 2024
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pressure vessel
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Sept. 05, at 8:48 a.m., while preparing for the scheduled PSV inspection there was a release of crude oil and gas from the desalter.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 04 Oct 2024
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On October 4, 2024, during walk-around a gas leak was detected (by sound) on the PSV line of the 2nd stage degasser/desalter. There was no activation of gas detectors, no plant shutdown and evacuation of personnel. Desalter was stopped, isolated and depressurized. This section of the pipeline belongs to the dead leg zone, which leads to the formation of deposits in it. The presence of deposits leads to general corrosion and thinning of the metal.

WHAT WENT WRONG?

- This section of the pipeline belongs to the dead leg zone, which leads to the formation of deposits in it. The presence of deposits leads to general corrosion and thinning of the metal on the internal surface of the pipe.

- External corrosion under insulation due to breakdown of the protective coating.
- Contributory factor: rough surface of fillet welding of piping support renders the imperfection of the external coating application.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review and update the specific line (and sisters) dead leg inspection work pack by absorbing the incident learning from metallurgical failure assessment.

To consider and take a decision on two options:

- to procure FLIR thermal imaging infrared cameras and prioritise high-risk areas to be inspected
- to introduce regular drone camera inspections

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 18 Sep 2024
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Contractor company was performing preparatory works for the repair of pipeline related to replacement of its defective part. During pilot hole drilling the gas condensate mixture leaked to the atmosphere from the pipeline of a well. The pipeline was depressurized to Unit 3 flare and to the horizontal flare of the well. No injured persons. >25 kg H_2S was released.

WHAT WENT WRONG?

- Inaccurate Pipeline Data and Documentation: For all Phase I pipelines, there is no reliable data available that can be used to identify accurately the pipelines routing, no as-built drawings, and the general map is incorrect and includes a lot of discrepancies. All Phase II pipelines have an accurate and reliable data and doesn't require special measurements.
- Underestimation of Conditions at Unit 3: Unit-3 is the oldest facility, and system and data in place (marks and signs are not available to indicate designated pipeline) are not designed to eliminate incidents. Always specific MS with additional measurements shall be adopted for any activities that are being executed at Unit-3 facilities and networks.
- Use of Unofficial Resources for Pipeline Identification: PIA used Google map and Field General Map (FGM) to define the pipeline location, the Google map is not an official document. Only one official source of information shall be allowed to be used for the entire organization. Master file is owned by CAD team at PED and update is issued once a year. Content for Phase I is known to be unreliable.
- Non-operational Cathodic Protection (CP) System: CP system is not operational, and there is no isolation between the underground pipelines. Availability of the CP system prevents corrosion and could support to identify the pipeline easily using particular measurements.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review Procedure "Preparation for Repair Work on Infield Pipelines" to define necessary control measures for Phase I pipelines.

- Implement additional control methods to prevent incidents during control hole drilling. These methods should ensure precise pipeline identification.
- Investigate if there is any equipment or technology that could help to identify the status of the pipeline.
- Revise the design of Cathodic Protection system operational for Phase I and the possibility to segregate CP for each well.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 08 Jan 2024
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Contractor employees were removing the DGS fixing segments for their replacement. 3 out of 4 fixing segments were removed. While removing the 4th segment, the DGS displaced from the compressor seat. The DGS struck the hand of one worker and tangentially grazed the other worker's finger. Both workers visited Unit-2 Medical Sick Bay where they were rendered the first aid. After that, one worker was sent to hospital for X-ray and further examination. Another worker returned to his job duties. Case under investigation – further information will be provided. Classification shall be clarified.

WHAT WENT WRONG?

- Use of not rated spade during positive mechanical isolation.
- Vent valve 109 was open but RB X 211 (manual) which is supposed to be used to control the pressure was Closed.
- Hazard of pressure accumulation in the 2-stage compressor was not considered in the risk assessment.
- Poor communication between involved parties during concurrent operations.
- Stop work authority was not applied by PRA (PTW was not withdrawn).
- Inadequate pressure control by Control room during N2 warm up activities.
- Excessive pressure builds up in 1st stage compressor above 15 bar.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Reinforce prohibition for all non-appropriately pressure rated spade/flange and not made from material resistant to the system content. Self fabricated metal spade/blinds to be withdrawn from use.
- Review the overall Isolation procedure, with particular focus on the process of spade selection, storage and installation as well as the handling and identification tag.

- Develop DGS warm up procedure with clear safety requirements for pressure control during warm up activities and Indicate PI (pressure gauge) requirement for N2 supply. Roll out to be conducted with all personnel involved in these activities including business partners.
- Temporary equipment procedure to be reviewed and updated with main focus on the measures to be taken before, during and after the period temporary equipment is connected to the process. Note: Management of pressure control to be considered.
- Identify and retrain relevant personnel on:

1. Mechanical Isolation procedure and how to properly apply it (integrity check properly done; All isolation points are properly assessed; rated spades are always used, etc.).
2. Safe isolation PS fundamental Rule #5 (We maintain safe isolation).
3. PTW procedure specifically for PRA PA to make sure work sites checks are properly done and work site is actually safe to start work RAMS are properly prepared and checked.
   - Ensure safety stand-downs are properly carried for all units and the lessons learnt are promptly cascaded to the key staff.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 02 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
NUMBER OF DEATHS: 1
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

An operation involving the intermittent injection of oil into the well casing for periodic cleaning the pump compressor pipes, and the filtration zone off mechanical impurities was carried out on the deck of an offshore platform. Following this operation, a fire and explosion occurred due to ingress of gas from the well into the enclosed space of the fixed cementing truck on the platform. As a result of the explosion, the well cementing truck operator sustained severe injuries and died on the spot.

WHAT WENT WRONG?

Lack of supervision of the valve opening sequence procedure.

Inadequate work planning.

Inadequate communication.

Inadequate process safety management procedure.

Inadequate control of work and risk assessment procedure.

Lack of competency.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Control of work procedure to be enforced.

ATEX certified equipment to be provided for hazardous works.

Process safety management procedure to be enforced.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 21 Oct 2024
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Prior to the commissioning of the well, a leak was observed on the wellhead.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 01 Nov 2024
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Drilling
MODE OF OPERATION: Drilling
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During isolation drilling, a variation in levels in the pool was observed (gain/loss), the screen was checked and a clogged box was observed, producing an increase in level in the separator and subsequent spillage of oil-based sludge through the discharge line (8") to the burn pit. The Separator is located upstream and does not have level sensors. LOPC: 10 m3 (3.2 m3 in pit and 6.8 m3 in screen containers).

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

DATE: 10 Jan 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

Operator found gas leak through orifice plate holder.

WHAT WENT WRONG?

Erosion due to sand content higher than expected during design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Orifice plate holder replacement plan. Improve design to mitigate erosion impact. Improve procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Assurance, review and improvement

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 29 Feb 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

At 22:00 on February 29, while the electro submersible pump was being started, an uncontrolled release of gas and subsequent blowout occurred in the wellhead.

The emergency response team controlled the event and no employee or contractor were injured.

Neither media nor external agencies involved.

WHAT WENT WRONG?

The classification of well in Drilling: Primary Producer – Initiates Surgent and Proposal for Completion was: Secondary Producer.

The proposed extraction method involved equipment not enabled to receive high pressure.

The selection of the extraction system was sent to Contractor before the test of F1, F2 and F3 (*) in which gas and pressure in BoP were determined.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Opportunities for improvement in the level of competencies to manage well barrier envelopes are evident.

Action Plan: Establish a procedure that defines the technical authorities and their competences in order to validate the extraction system and well barrier envelopes based on the associated risk.

BARRIERS:

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 07 Aug 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

During the operational round, loss of containment was detected in the discharge line of the V-202 separator.

WHAT WENT WRONG?

Erosion by sand.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improve design to mitigate erosion impact.

Improve procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 24 Nov 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Loss of gas containment due to failure of a gas line valve stem.

WHAT WENT WRONG?

Material Failure

Deficiencies in supplier quality control

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improve the quality assurance system

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Assurance, review and improvement

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 17 May 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

The ingress of a hydrocarbon slug from the producing wells caused the levels in the treatment section vessels to rise and flood. As a result, unstabilised condensate entered the glycol regeneration train. Finally, the equipment in the glycol regeneration section was flooded and oil was vented through the vent stack. This activated the flammable gas detectors in the surrounding areas and caused the unit to shut down.

WHAT WENT WRONG?

The need to simultaneously bypass multiple layers of protection due to configuration/calibration problems after the initial planned maintenance of the unit, combined with poor risk management, allowed the undetected overfilling of various vessels until condensate venting occurred through the stack of the glycol regeneration section.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

In the case of safety critical equipment inhibition, it is essential to take a holistic view in order to identify multiple inhibits and assess the overall risk level for decision making.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 21 Mar 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Through a routine tour, the operator detected a hydrocarbon spill coming from the delivery pump, which was caused by a rupture of the packing press. At the time of discovery, the operation was normal with a pressure of 32kg/cm2 in the pump, which after detection was corroborated by PI some parameter deviations in pump rpm and sump level.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 May 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pig launcher/receiver
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Broken ½" nipple at the chemical product (paraffin inhibitor) dosing point at the junction with the pipeline, Reception Trap Zone Section 2 (8" steel), within the pumping station.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 May 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

HC spill was detected by pitting on the control line.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

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DATE: 23 May 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Plant operator, on his routine tank tour, observed a problem in the tank discharge trunk line inside the bund of a tank.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment.

DATE: 29 May 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

In order to commission the TK-118, a fresh water cushion must be added to cover the sacrificial anodes. In the first instance, it had been planned to use the connection corresponding to the washing water (which is linked to the TK-111, TK-113, TK-118 and Free Waters tanks to take advantage of its temperature, but it was frozen. For this reason the line was disconnected to carry out the filling with another temporary connection. Given the increase in temperature the day before, the disconnected line began to drain, causing the spill of product in the line and the discharge of oil from Free Waters.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 04 Jun 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Overflow of separator through manhole. By Pass valve was found half open.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 07 Jun 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil spill detected due to breakage in backpressure valve.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

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DATE: 14 Jun 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

After releasing the packer, the single pipe is removed, 33 pipes are removed, a manoeuvre valve is placed and the well level is completed between columns with 600 litres of fluid (flowback water). When removing the valve to continue removing pipe, direct upwelling of the well is observed. Team personnel places a manoeuvre valve in the pipe coupler at the wellhead (with difficulty due to fluid leakage). After several attempts, the valve was placed and the well was controlled.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

DATE: 08 Jul 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During a routine tour, the scout detected an oil leak emerging from the ground on the property inside and outside the site fence. Role was activated and activities were carried out to divert production, identify and take out of service the line that had the fault. Once the production bypass had been carried out, the line was closed and contingency controlled. Background; In February 2021, an environmental incident occurred in the same pipeline, in which it was investigated and had an integrity report on the analysis of the Root Cause of the Failures and from which actions derive that will be linked to this new event.

Actions detail:

- Secure resources and deliver on all mechanical cleanup tasks.
- Review system-wide mitigation plan and improve alert communication.
- Characterize fluid and issue mitigation plan to avoid deposition of paraffins and inorganic scales.
- Plan and execute inline inspection along the entire length of the pipeline.
- Perform mechanical cleaning of the pipeline immediately.
- Perform permanent repair of the pipeline.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

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DATE: 11 Jul 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gas leak through water discharge line of rental separator. Broken studs are observed in the 3" s600 ball valve body.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

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DATE: 31 Jul 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Upset
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Major incident occurs due to burst plate rupture in general two-phase separator.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Emergency Response Equipment and Systems

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

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DATE: 09 Aug 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Temporary
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

During the Well Testing operation of the well, the threaded end of the ball valve and the nipple piping to the separator were detached.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 18 Aug 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Loss on battery trunk pipeline at 900 meters is identified with our own personnel of the installation (approx.), on pipe trace. The spilled volume is mixed with water accumulated in a shallow area due to thawing after a climatic contingency.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 27 Aug 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

As a result of an electrical problem, the production water pumping system stopped, causing a high water level in skimmer. As a mitigation measure, raw production was derived from the skimmer contingency tank; reaching the high level, plant closure and product spill due to overflow leg.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 05 Sep 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

The operator proceeded to start loading, then descended from the loading ramp to check tractor levels, when he noticed the tanker overflowing.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 06 Sep 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

On their tour operators observed an open pump discharge valve which had been opened in the previous trip.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 11 Sep 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The runner detected a hydrocarbon leak in the Ø 8" line at the outlet of the collector. Four wells were closed, isolating and depressurizing the affected line section.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 17 Sep 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

On the well pad, pumping was enabled at 18:03, leaving the pipeline that enters the collector. At 7:40 p.m., a spill was detected because the valve between the receiver and the collector was "closed," generating the passage of crude oil to the receiving trap, and since the purges were open, a spill was generated due to the overflow of the chamber.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 09 Oct 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

There was a major loss of crude oil containment, due to the breakage of the conduction line of a well, 50 meters from another well.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 28 Dec 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Loss of Containment of the Gas Pipeline: During normal road review operation with a motor grader, the tip of the shovel touched the gas pipeline. A loss of containment occurred, causing a stone to hit the window of the motor grader, breaking the glass. Without consequences for the operator. Immediately the machine operator retreated to the main road intersection at Battery, to move away from the exposure area.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 12 Nov 2024
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Loss of hydrocarbon containment occurs due to overflow of drainage chambers when the purge valve was found open in the discharge line of the Booster pump, during start-up.

WHAT WENT WRONG?

N/A

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 08 Jan 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil spill due to a hole in the section of the production flow line from the production wells to the oil gathering station. The phases oil, water, and gas were present as the line carries the raw production from the wells to the station.

WHAT WENT WRONG?

Lack of clear guidance in standards regarding the identification of inspection points in areas susceptible to corrosion-erosion. There are no alerts in the inspections system for cases of adding new inspection points to the system. There is a lack of guidance in standards on critical analysis for new measurement points in older operational pipelines. Inspection procedures do not define applicable techniques for erosion cases. Insufficient resources for conducting the required scope.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review the local standard with guidelines for identifying inspection points in areas susceptible to corrosion-erosion and include the requirement to document the selection of these points. Conduct an evaluation of similar lines to prevent the omission of inspections in areas susceptible to corrosion-erosion. Create alerts in the inspection system for new monitoring points, comparing corrosion rates with nominal thickness and/or with other measurements in the same section.

Revise the local standard to incorporate a critical analysis method for new measurement points in older lines.

Assess the best technique and revise the local standard to include the most appropriate method for verifying the integrity of sections susceptible to erosion.

Establish criteria for prioritizing the scope and treatment of actions based on the criticality of occurrences.

Provide resources to complete inspections at the Separation Stations in a timely manner.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

NONE: Unspecified

DATE: 06 Feb 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

The incident involved a failure during a routine maintenance operation at an oil treatment station. The maintenance was focused on the temperature control valve of the oil treatment unit. During the removal of the temperature sensor from the thermowell, the thermowell was dislodged, leading to a release of oily water at high pressure and temperature. This initial leak was not immediately recognized, and the situation escalated when oil began to leak from the unit, eventually leading to the release of gas that ignited, causing a significant fire.

WHAT WENT WRONG?

- Intervention during the maintenance of the temperature control valve. Specifically, the thermowell detached during the removal of the sensor, leading to a loss of containment of oil and water.
- Non-existent emergency procedures: the procedures in place for emergency situations were inadequate. The adopted emergency procedure did not properly address the need for immediate containment actions, which resulted in the intentional drainage of fluids into open channels, exacerbating the situation.
- Inadequate risk assessment: there was a lack of proper risk assessment regarding the actions taken during the emergency. The team did not adequately evaluate the consequences of draining fluids, which led to the release of gas and subsequent ignition.
- Inadequate procedures: the operational procedures did not provide clear guidance for emergency situations, particularly regarding the need to depressurize the equipment before emergency actions.
- Inadequate maintenance planning: the maintenance plan did not include necessary checks for the integrity of the thermowell, which contributed to its failure during the maintenance operation.
- Inadequate project requirements: the design and installation of the equipment did not meet the necessary safety standards, particularly regarding the pilot light system, which remained lit during the emergency, contributing to the ignition of the released gas.
- These root causes highlight the need for improved operational procedures, better risk assessment practices, and enhanced maintenance protocols to prevent similar incidents in the future.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Include a checklist for torque and tightening of instruments and thermowells in operational and maintenance standards.
- Add a task for checking the threading of equipment before maintenance release in the maintenance procedures.
- Review and update the operational procedure for the oil treatment system, ensuring it addresses emergency situations effectively.
- Improve the integrity of the ignition system considering the area's risk classification.
- Assess and correct the logic for emergency shutdowns, including pilot lights.
- Update documentation related to the furnace and remove obsolete documents.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

NONE: Unspecified

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 08 Jan 2024
COUNTRY: Brazil
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

During the well damping operation by bullheading, the pumping pressure reached the configured limit of the relief valve (hydraulic pop-off) of the mud pump, causing its activation and consequently the failure of the primary Well Barrier Envelope. The opening of this valve was not noticed by the team on board and resulted in the loss of hydraulic pressure on the formation. Furthermore, as monitoring parameters were not fully established, as well as a hidden failure in monitoring the pop-off discharge tank, there was a short delay in closing the valves of the subsea workover assembly and Surface Flow Tree (secondary Well barrier envelope), allowing the migration of gas through the production column and DPR (Drill Pipe Riser), reaching the tank room (100% LEL) through the relief valve and, subsequently, discharge on the main deck of the unit through the air exhaust system of the storage room tanks, covering a significant extension of unclassified area (peak of 64% LEL on the sensor close to the Well Testing plant).

WHAT WENT WRONG?

- Inadequate operating procedure: There was a failure to monitor parameters related to the fluid during the well damping operation.
- Failure in the management and control of critical operational safety elements: There was a late execution of the well closure.

- Failure in the risk identification and analysis methodology: Exhaust from the tank room directed to an unclassified area close to the well test boiler.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Include in the Driller's checklist the verification that the tank sensors are activated.
- Communicate to the teams that during the pumping of fluid, the level of the tank used for pumping must be monitored, that is, tank that must lose the pumped volume, as well as the tank that receives the fluid in case of opening the pop-off, and consequently will gain volume in case of opening it.
- Include audible and visual alarm in the driller cabin in case of pop-off valve opening.
- Include questions in the design and operation checklists related to pumping pressure limits in BH (bullheading) operations in LWO (light workover).
- Establish critical operations for on-site monitoring by inspectors, toolpusher and others involved in hydrocarbon risk situations. (Ex: Home well damping, dry test execution, and others).
- Establish a list of loss of containment evidence for LWO that result in the need for immediate closure of the well. Ex: 1-general alarm of CH4 of unknown origin; 2-Minimum pressure limit on the head at the beginning/during BH; 3-Unexpected return of hydrocarbons at the Well Testing plant and others.

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: 21 Feb 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During offloading operation, the Pumpman noticed the loss of primary containment originating from a hole in the crude oil pump #2 suction line located in the pump room. Without delay, he promptly communicated the exact location of the LOPC to the Marine Control Room Operator (MCRO) who was on duty at the time. The MCRO swiftly initiated a halt to the ongoing offloading operations. The leaking volume was naturally routed to the channels and bilges.

WHAT WENT WRONG?

Internal corrosion in the related line.

Inspection did not detect the internal corrosion on the related pipe.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Use inspection tools in order to verify the thickness measurements on the pipewall (internal and external corrosion) and identify area of material loss.

Evaluate to perform an Advanced NDT (PECA) in order to have a more robust understanding of the pumproom piping condition – by using a extraordinary inspection schedule plan.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 04 Apr 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

An emergency shutdown was activated due to the detection of a fuel gas leak. The leak was identified as originating from the rupture of a diaphragm seal on a differential pressure gauge in the fuel gas scrubber. The gas was detected at 6:05 AM, triggering the activation of emergency measures. During the incident, there were no injuries, but the situation resulted in a loss of containment of flammable gas. Production resumed the following day at 16:41. The lack of adequate training and team preparation may have impacted the response capability and the execution of safety procedures.

WHAT WENT WRONG?

The incident resulted from a combination of factors, including inadequate personnel qualification and poor team communication, which may have affected the response and decision-making. Established standards and procedures were not properly applied, and equipment design did not fully comply with required specifications, partly due to insufficient consideration of environmental factors. Additionally, certain failures were deemed tolerable despite their operational impact, and corrective actions taken after the incident were insufficient to fully address the underlying issues, highlighting the need for further improvements in process safety.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The investigation highlighted several key areas for improvement. It is recommended to replace the current material used for the diaphragm seal bolts to prevent corrosion and to implement measures that inhibit crack formation, which can contribute to further deterioration. Additionally, the study calls for identifying the number of instruments from same fabricant installed across similar Offshore units. To further mitigate corrosion, it is advised to isolate the affected instruments and apply industrial grease and/or silicone, along with conducting biannual inspections. Furthermore, the group's technical standards should be revised to clearly define the requirements for the diaphragm seal, and alerts should be issued to the entire fleet and project teams regarding the identified failures and necessary corrective actions.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 27 Apr 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

During normal platform operations, the gas detector in the module triggered an alarm. A production operator, using a portable multi-gas detector, confirmed a gas leak in the inlet line of a Gas Separator vessel. It was decided to immediately and safely shut down production for intervention.

WHAT WENT WRONG?

It was identified that the failure mode of the event was due to the material being carbon steel (inadequate design) combined with the change in interaction with the material due to alterations in the fluid, and carrying of solids. Additionally, a human cause was also observed in the decision for a controlled shutdown instead of an emergency shutdown, which over time led to a Tier 1 event, occurring due to the choice being a human decision.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Recommendations for equipment problem encountered (erosion source), and review the decision-making process for plant shutdown.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 04 May 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

During the opening of the pipeline for the replacement of the rupture disc of the Low-Pressure Flare system, the release of high-temperature combustion products (flash) occurred, resulting in localized burns on the hand and leg of two workers performing the activity.

The process plant was in a total shutdown, partially depressurized, and the section to be intervened was blocked and cleared for work. The rupture disc bypass valve remained open to allow the release of any residual gas and was aligned with continuous purging using natural gas. However, the undetected partial opening of a valve allowed oxygen to enter the pipeline, contributing to the incident.

WHAT WENT WRONG?

The workplace layout had ergonomic inadequacies, particularly in the positioning of controls and displays, making it difficult to see the indicator arrow on the valve reducer and, consequently, confirm its physical position. Additionally, there was a failure in safety across the design, construction, installation, and decommissioning phases, as evidenced by the incorrect installation of the instrument during construction.

The operational procedure also had gaps in the hazardous energy control system. These gaps included the absence of a requirement for prior verification of the position indicator for manual valves, a lack of guidance for confirming closure using the reducer arrow, and no clear guidelines for assessing the feasibility of reducing blockage.

The system itself proved intolerant to errors, the platform management system, does not require the inclusion of the specific procedure.

Finally, the intervention planning was inadequate, as the plant was partially depressurized, which required the Flare pilot to remain lit.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

To enhance safety and operational reliability, several recommendations have been proposed. First, temporary access should be provided to ensure clear visibility of valve position indicators during operations and maintenance. Additionally, when accepting new units, it is important to verify that the indicated position of control devices matches the actual position of the valves.

Another recommendation is to improve operational procedures to ensure that valve position indicators are always correctly aligned and that closure confirmation is performed directly on the valve in the field.

For activities involving line openings in the Flare system, it must be ensured that the valves are properly blocked and that a safe isolation process is in place. If full isolation is not possible, the entire section of the system that may release flammable gases must be completely depressurized, the Flare pilot must be turned off, and the pipeline between the opening point and the system's endpoint must be filled with inert gas.

Additionally, it is essential to improve the criteria for verifying the sealing integrity of valves used for blocking, ensuring they are effectively closed before starting activities. It has also been recommended that this type of verification be incorporated into operational management systems, increasing traceability and control over these measures.

Finally, the need to ensure that, in any intervention involving line opening in the Flare system with the pilot lit, valve blocking is confirmed, and continuous purging with inert gas is carried out, has been reinforced, significantly reducing the risk of incidents.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

DATE: 10 May 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Filter
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

During the transfer of dead oil with a transfer pump, gas was detected in the pump room, triggering a general alarm. After ruling out the presence of gas, the emergency response team entered the pump room and identified an oil leak in the filter of that pump.

WHAT WENT WRONG?

Internal corrosion of the line due to the circulation of produced water, which was not in the equipment design, led to accelerated corrosion of the system. Another point was the incorrect classification of the equipment in the categorization for inspection and maintenance, resulting in its deprioritization during inspection management. Additionally, the equipment was not isolated from the process during fluid manoeuvres of other pumps, leading to unnecessary exposure of the equipment that was not involved in the operation. Another finding was that there were several Tier 3 events in the system that did not receive preventive actions, only corrective actions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The recommendations from the investigation of the incident on the FPSO include creating and implementing a guidance for reporting process safety events, reinforcing the use of tool features for reporting repetitive events, and verifying unnecessary equipment exposure during transfer manoeuvres while addressing studies for ALARP and necessary actions. Additionally, there should be engineered solutions for the equipment's work environment, including replacement or internal coating, as well as an assessment of the cargo system for points needing coating and corresponding preventive/corrective actions based on findings. Finally, a review of the procedure is needed to enhance the requirements for the classification and prioritization of equipment in inspections.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

NONE: Unspecified

DATE: 21 May 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Occurrence of emergency shutdown caused by confirmed gas in module of gas exportation, due to inadvertent opening of an entry hatch of the cargo tank.

WHAT WENT WRONG?

The recommendations regarding the basic causes of the incident on the FPSO highlight several root causes: the documentation presented by the execution team lacked information related to interventions on the tank entry hatches, hindering proper planning and execution of the work, in violation of the permission of work procedure. The procedure for the risk analysis for the task was found unclear whether an isolated intervention should follow an initial matrix or require a new one, leading to ambiguity. Additionally, there was no identified painting/inspection plan for the tanks that included the painting of the hatch tags, and the committee noted a lack of standardization regarding the replacement of entry hatch gaskets, which allowed for the reuse of materials. There was also an absence of standard instructions for signalling equipment under intervention, and no records in deviation and anomaly reporting tools confirmed that the hatches were swapped between tanks and no technical documentation demonstrating that.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The recommendations from the investigation of the incident on the FPSO include conducting a workshop with the contracted company to assess the approval of guidelines provided by contractors to ensure compliance with the Work Permit standard. Teams involved in planning should be guided on the signalling of 'Critical Service' for activities involving tank openings, ensuring that these activities are properly integrated into the management system to facilitate a more thorough and detailed risk analysis. Additionally, the Inspection Plan for cargo tanks should be reviewed to include the verification of the condition of paint identification on the entry hatches and domes for structural tanks, comparing it with documentation. Lastly, a campaign should be structured to encourage the reporting of identified TAG deviations on the oil platform.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 12 Jul 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

During the process of returning the portside drainage system tank to operation, which had been inactive for months, the stability technician responsible for the control room authorized the intervention in the Pump House using a continuous Work Permit (WP) for the unblinding of the portside tank. However, the activity performed did not match the scope of the WP. Instead, a valve was removed from a starboard tank and installed in the portside tank, resulting in a loss of containment at the portside flange, which had been opened for this installation.

The continuous WP had been initiated on 08 July 2024 for the unblinding of the tank, and the loss of containment occurred on 12 July 2024. Additionally, it was found that the portside tank's isolation had already been altered to allow pressurization with inert gas prior to the loss of containment as part of a tightness test.

WHAT WENT WRONG?

The lack of prior analysis of safety conditions and risks in the workplace resulted in the failure to consider the risk of fluid passage through the valves during activity planning. Additionally, it was found that the intervention took place on two different pieces of equipment using a single Work Permit (WP). The adopted isolation plan was inconsistent with established guidelines, which required "positive isolation" due to the high risk factor; however, the work was authorized with a simple, unverified lockout. The absence of a specific procedure for the activity was also identified, preventing a documented sequence of manoeuvres for the release of the drainage tanks (slops) for opening or inspection, as well as a clear definition of the necessary lockouts and isolations to ensure operational safety. As a contributing factor, damage to the sealing seat of the butterfly valve occurred due to improper torque applied to its fasteners, compromising its sealing efficiency.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

It is essential to ensure strict compliance with the recommendations for the use of the continuing Work Permit (WP), verifying daily that the ongoing activity matches the one originally evaluated in the hazard analysis. The task description in the WP should be as detailed and compatible as possible with the actual work to be performed. Whenever it is identified that the activity being carried out is not fully covered by the WP, a new preliminary hazard assessment and a new WP should be issued, and no new tasks should be added to an ongoing WP.

The activities that can be released by the control room technicians aboard the vessel should be evaluated, with a particular focus on those classified as high risk and requiring closer monitoring. Additionally, it is necessary to ensure that maintenance, assembly, and operation personnel are aware of the specific recommendations and guidelines for the assembly of valves, ensuring proper installation and that functionality tests are carried out in accordance with the established guidelines to avoid operational failures.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

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DATE: 14 Jul 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Loss of containment of 27 m^3 of oily water (Water phase: 22,140 kg + Oil phase: 4,323 kg = Total: 26,463 kg), through the top valve of the port Slop tank of the FPSO, during an operation to receive oily water resulting from the backwashing of the offloading hoses.

WHAT WENT WRONG?

The port slop tank overflow was caused by a combination of factors. First, the high-high level alarm set point was incorrect, allowing the tank to overflow without the alarm being activated. In addition, there were intermittent failures in the high-level sensor, which did not function when required. The investigation showed a history of spurious activation of this alarm, as well as revealed that the contractor did not implement a decision-making criterion in cases of reaching critical levels of the slop tanks, following a similar accident in 2022, which could have prevented the occurrence. Finally, the control methodology was flawed, as there was no robust system to ensure that the dump valve remained closed, in deviation from the guidelines of the operational procedure, which could have mitigated the overflow.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Correct the virtual high-level alarm set for the port and starboard slop tank to a level below the drop line nozzle with a lock for manual adjustment. Correct the set point for the very high-level alarms for the port and starboard slop vessels, keeping it below the drop line nozzle level. Perform maintenance on the port slop high-level alarm, ensuring its operability. Perform tests on the other level alarms for the tanks and vessels in the open drainage system, and perform maintenance on the sensors that are not fully operational. Create a maintenance plan for calibrating the level sensor for the open drainage tanks. Review the related operating procedure and train the team on the updated version. Implement a reliable system for controlling the opening and closing of the dump valves, with leadership signatures.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

NONE: Unspecified

DATE: 14 Jul 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Loss of primary containment of the flow of a mixture of flammable gases, through the connection between vertical connection module-A and the Funnel. It was not possible to determine the worst moment of the leak, considering the total leak time of 17.2 hours and the release rate of 4988.37 kg/h. During the inspection carried out on the set, intense hydrate formation was evidenced in the vertical connection module-A equipment and a misalignment was noted between the valve panel and the X-mas Tree HUB, indicating a possible disconnection of vertical connection module-A from the funnel.

WHAT WENT WRONG?

Failure to apply hazard identification mechanisms and/or risk analysis prior to implementing the modification. The design change was unable to anticipate the formation of a chamber by adding a seal. Failure to monitor and evaluate the results of inspections and tests. The vertical connection module visual locking indicator did not reach its full displacement after the locking procedure. No qualitative or quantitative risk identification and analysis. The equipment manufacturer did not identify design documentation regarding the identification of risks and failure modes of the equipment. At the time of supply, the study of failure modes was not a requirement of the Technical Specification, which was incorporated into subsequent projects.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Implement and demonstrate a process and methodology for analyzing and quantifying the risk of changes in projects, covering other manufacturers of subsea equipment. Reevaluate all inspections and installation reports of vertical connection modules in the scope list and, in case of signs of gas leakage (bubbling or hydrate) and/or locking indicator out of position, contact the integrity manager to perform a risk analysis for operational continuity. Assess the need for differentiated inspections. Provide in the procedure that the hydrostatic test should not be used to validate the locking of vertical connection modules. Review the Technical Specification. Carry out the collection of vertical connection module information. Coordinate, together with the operational area, the technical detailing and execution of the dissection of the vertical connection module to verify the sealing components and locking system. Re-evaluate the fault tree based on the dissection results and propose additional recommendations, if applicable.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Stakeholders and customers

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 16 Oct 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Loss of primary containment and emergency shutdown due to gas detection confirmed, after a failure in the tubing of pressure transmitter of the slop pump discharge. It was found that the vibration dampers of the slop pump discharges were inadequately calibrated, losing their ability to absorb vibration. As a result, the pressure transmitter fixing bolts broke, leaving it in a swing, causing a consequent failure in the tubing.

WHAT WENT WRONG?

The plans and procedures for inspection, testing and maintenance are inadequate. The pulsation dampeners were not calibrated properly due to the existing maintenance plan being incorrect. Manufacturer's recommendation and/or standard and/or standards for inspection, testing and maintenance procedures not implemented. The original bracket was not painted when required and its failure caused accumulation of damage in the tubing, contributing to its future failure. Failure to implement the technical report recommendation regarding the calibration range of the pulsation dampeners.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review the maintenance plan, changing the pre-charge pressure of the dampers to the range of the operating pressure according to the technical report. Prioritize the mechanical treatment and painting of the instrument support structures. Issue a technical alert to the change management coordinators, focusing on verifying whether the evidence of compliance with the Risk Analysis recommendations is consistent.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

NONE: Unspecified

DATE: 29 Nov 2024
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

683,5 Kg under investigation. During the simultaneous operation between the accommodation for support in maintenance and safety unit and the FPSO, there was an incidence of a gas plume from the Vent Post of the production unit, which is responsible for depressurizing the cargo tanks and Slop tanks. This caused the flammable gas sensors of the support in maintenance and safety unit to be triggered at elevated concentration levels, activating the safety logic and triggering the confirmed gas alarm. As a result, a programmed disconnection was immediately commanded according to the emergency scenario.

WHAT WENT WRONG?

The root causes identified in the investigation of the incident on the FPSO included conflicting requests and undefined parameters for safely executing the connection/reconnection manoeuvre of the maintenance and safety unit, particularly concerning the operation of the vents. Due to critical weather conditions affecting plant operations, it was not possible to maintain continuous depressurization as mandated by the standards. Consequently, adjustments to production were made to comply with the criteria set forth in a procedure without adequately addressing the prevention of gas plumes in the maintenance and safety unit. Furthermore, not all stakeholders were aware of the necessary parameters for decision-making related to the connection/reconnection process.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The recommendations from the investigation of the incident on the FPSO include the implementation of a panel with an algorithm for analyzing process parameters and environmental conditions to determine the connection and disconnection status of the maintenance and safety unit. A systematic decision-making process should be established for evaluating operational parameters and environmental conditions through meetings involving the maintenance and safety unit, the platform, and the onshore team to facilitate simultaneous operations. Additionally, it is recommended to incorporate and disseminate all current parameters in the management system to aid in decision-making regarding the connection/reconnection of the maintenance and safety unit.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

FATAL INCIDENTS CLASSIFIED AS TIER 1 PSE

In this Section, fatal incidents reported in the safety performance dataset that were also reported as Tier 1 PSE in the process safety data set are presented.

ASIA / AUSTRALASIA OFFSHORE

DATE: 02 Aug 2024
COUNTRY: Malaysia
NUMBER OF DEATHS: 2
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Engineer, Scientist, Technician, **Body part:** Respiratory system, **Nature of injury**: Burn, **Time in service:** > 1 year < 5 years

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Other, **Body part:** Respiratory system, **Nature of injury:** Burn, **Time in service:** 0 months < 3 months

NARRATIVE:

Cargo pump explosion aboard platform supply vessel while performing liquid transfer activity. Seven personnel sustained injuries. Two deceased persons (DPs) and five injured persons (IPs).

WHAT WENT WRONG:

Under investigation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Under investigation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 02 Sep 2024
COUNTRY: Azerbaijan
NUMBER OF DEATHS: 1
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Hot work
SECONDARY LIFE-SAVING RULE: Bypassing safety controls

FATALITY

Function: Production, **Employer:** Company, **Occupation:** Process/equipment operator, **Body part:** Unspecified, **Nature of injury:** Major/multiple system trauma, **Time in service:** 10 + years

NARRATIVE:

An operation involving the intermittent injection of oil into the well casing for periodic cleaning the pump compressor pipes, and the filtration zone of mechanical impurities was carried out on the deck of an offshore platform. Following this operation, a fire and explosion occurred due to ingress of gas from the well into the enclosed space of the fixed cementing truck on the platform. As a result of the explosion, the well cementing truck operator sustained severe injuries and died on the spot. Investigation found out that gas ingress occurred due to violation of the valve opening sequence procedure.

WHAT WENT WRONG:

- Inadequate control of work system and risk assessment
- Inadequate design.
- Lack of supervision.
- Failure of work area preparation for fire hazardous operation.
- Inadequate hazard communication.
- Inadequate process safety management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Risk-taking behaviour can lead to serious accidents.
- The hot work permit system should be implemented, requiring gas testing to be conducted in hazardous areas.
- ATEX certified equipment to be used in hazardous areas.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

FATAL TIER 1 PSE NOT INCLUDED IN THE SPECIFIC PSE DATA COLLECTION

Participation in the IOGP annual collection of safety and process safety performance data is voluntary, and not all Member Companies participate in each specific data set. In addition to the Tier 1 PSE data presented above, a further 3 Tier 1 PSE with fatal outcomes were reported as part of the 2024 safety performance data set (IOGP Reports 2024s and 2024sf). Narrative descriptions for these additional Tier 1 fatal PSE have been included in this Section of this Report for the purposes of learning. They are not included in the data analysis or performance metrics within Report 2024p, as the reporting companies did not participate in the specific PSE data collection.

AFRICA OFFSHORE

DATE: 20 Mar 2024
COUNTRY: Gabon
NUMBER OF DEATHS: 6
FUNCTION: Unspecified
CAUSE: Explosion, fire or burns
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

FATALITY

Function: Drilling, **Employer:** Company, **Occupation:** Foreman, Supervisor, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part**: Internal organs, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Foreman, Supervisor, **Body part:** Internal organs, **Nature of injury:** Unspecified, **Time in service:** –

NARRATIVE:

Ignition and explosion during WO Well Control Incident.

WHAT WENT WRONG:

Well Control Incident and Fake ATEX certification on spot lights.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Vendor control policy revision.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

NORTH AMERICA ONSHORE

DATE: 07 Jun 2024
COUNTRY: Mexico
NUMBER OF DEATHS: 1
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

NARRATIVE:

Fire in large-scale exploitation operation.

WHAT WENT WRONG:

Lack of communication between workers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reinforcement of communication during start of activities meetings.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA OFFSHORE

DATE: 06 Apr 2024
COUNTRY: Mexico
NUMBER OF DEATHS: 3
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

NARRATIVE:

Fire in the processing centre.

Due to inadequate planning of works by the Command Line and failure to communicate to the operators the manual operation of the high-pressure fuel gas heaters due to high temperature, there was a sudden loss of containment and release of gas in the heater, causing a fire and explosion. The fire spread due to the storage of hazardous chemicals (toter) located on the second and third levels of the platform and the poor coordination in emergency response.

The root cause analysis to identify the causes that led to this accident and establish recommendations is under review.

WHAT WENT WRONG:

Lack of planning of activities.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Do not prioritize production over safety.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

FATAL INCIDENTS RELATED TO PROCESS SAFETY

In this Section, fatal incidents reported in the 2024 safety performance dataset that were related to process safety but were not classified as PSE are presented.

MIDDLE EAST OFFSHORE

DATE: 19 Feb 2024
COUNTRY: Qatar
NUMBER OF DEATHS: 1
FUNCTION: Production
CAUSE: Confined space
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Confined space
SECONDARY LIFE-SAVING RULE: Bypassing safety controls

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Respiratory system, **Nature of injury:** Inhalation, poisoning, intoxication, asphyxiation, drowning, **Time in service:** 0 months < 3 months

NARRATIVE:

The Injured Person (IP) was found unconscious with his head and shoulders inside an exchanger channel head which was oxygen deficient due to nitrogen purging. The IP's colleague observed the scene and promptly summoned help from a nearby Safety Officer, who in turn called for emergency services. The IP was retrieved from the exchanger channel head and first aid was initiated on site. The IP was then transported to Hospital where he remained in a critical condition and passed away on 28 February 2024.

WHAT WENT WRONG:

- Sustain Barriers: The IP entered a restricted area to perform an unplanned inspection.
- Respect Hazards: The IP exposed himself to an oxygen deficient atmosphere.
- Apply Procedures: The IP failed to:
  - Obtain the required authorisation and permits and establish its required controls.
  - Use Self-Contained Breathing Apparatus (SCBA) when entering the barricaded area with an oxygen deficient atmosphere.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Adequate planning and preparation allow identification of tasks, associated hazards, and implementation of controls to achieve safe and timely execution.
- It is important to consistently follow procedures and stop work if compliance becomes challenging. Speak up and raise concerns if procedure requirements are unclear, difficult to implement, or involve unavailable tools at site.
- To access confined spaces or its exclusion zone, a PTW is required along with the implementation of all associated controls.
- A nitrogen rich environment can rapidly displace oxygen, cause unconsciousness within seconds, and potentially cause death due to asphyxiation.
- Violating Life-Saving Rules can result in fatal consequences.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

HIGH POTENTIAL EVENTS CLASSIFIED AS TIER 1 PSE

In this Section, high potential events reported in the 2024 safety performance dataset that were also reported as Tier 1 PSE in the process safety data set are presented.

EUROPE ONSHORE

DATE: 03 Apr 2024
COUNTRY: Austria
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Blow out. During mounting, the borehole started to overflow.

The crew tried to secure the well – without success.

Alerting and mobilisation of the necessary emergency personnel.

After installing two preventers, the well was secured and pumped dead with 27m^3 of mud.

The well was regularly checked up overnight, and clean-up work was also carried out.

A Workover Rig was replaced, the well was additionally secured with a long stroke (LS) packer and the blowout preventer (BOP) system, including the Christmas tree lower part, was completely replaced.

WHAT WENT WRONG:

Insufficient fluid in the well bore.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Exact volume calculation of well bore, sufficient fluid on location.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 29 Oct 2024
COUNTRY: Iraq
FUNCTION: Production
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Hot oil release led to fire.

WHAT WENT WRONG:

The root cause of the incident is a leak in one of the hot oil convection tubes. This leak caused oil to drop onto the ignited burners and other hot surfaces, leading to the emission of black smoke and subsequent fire. The situation escalated, requiring manual Emergency Shutdown System (ESD) activation and the shutdown and depressurization.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Always follow the MoC process to meticulously record and evaluate the impacts of any changes in hardware or operations, such as temperature, pressure, composition, and flows.
2. For any maintenance, corrective maintenance (CM), or inspection work that is deferred or postponed, it is crucial to document, implement, and monitor clear mitigation measures and controls.
3. Encourage the Emergency Response Team to plan, drill, and exercise various scenarios relevant to the plant equipment and its associated hazards or risks.

CAUSAL FACTORS:

No causal factors allocated.

DATE: 22 Oct 2024
COUNTRY: Iraq
FUNCTION: Production
CAUSE: Unspecified – Other
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Liquid carryover at cold flare.

WHAT WENT WRONG:

The main root cause of the incident is the restriction in the piping to the burn pit. This restriction was caused by the U trap acting as a flame arrestor, which resulted in increased pressure in the piping header. The elevated pressure enabled the potential for reverse flow, ultimately leading to the issues and subsequent emergency response measures taken.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The main lesson learned is the importance of effective implementation and proactive monitoring of controls and procedures to ensure safety and prevent incidents. This includes following Company guidelines, conducting proper operator monitoring, implementing previous controls, and responding to abnormal conditions promptly.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST OFFSHORE

DATE: 28 Jun 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Electro-chlorinator unit cell cover blown off causing multiple injuries. Following a corrective maintenance activity carried out during the day, the Programmable Logic Controller (PLC) of the Electro-Chlorination Unit was reset. After the reset, a violent explosion occurred in the Electro-Chlorination Unit, causing a blast and severe injuries to the Injured Person (IP) who was in the immediate vicinity. The general platform alarm was initiated, and emergency protocols were followed. The IP was found lying on the deck. He was transferred to the clinic, and then safely evacuated to a Doha hospital by Medevac chopper.

WHAT WENT WRONG:

- Respect Hazards: Hydrogen detection was not adequately addressed in the current design.
- Apply Procedures: Operating and maintenance practices did not follow the recommendations from the Original Equipment Manufacturer (OEM). There were gaps in the application of Company procedures (e.g., Permit to Work).
- Watch for Weak Signals: There was a failure to apply executive action in response to the observed degradation of various systems (Management of Change, quality control, surveillance, emergency management, and the Permit to Work system).
- Maintain Competency: There was an absence of Hydrogen risk awareness among offshore operations personnel.
- anage Change: No Risk Assessment or HAZID/HAZOP was carried out within the framework of the eMOC, and a last-minute change was implemented without following the eMOC procedure.

What went well?: Emergency Response was immediately initiated, and Case Management was followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that all steps of the MOC process are effectively applied and that all expected deliverables are issued at the required level of quality.
- Verify that any Hydrogen risk is adequately addressed in the safety reviews of Electro-Chlorination units, including the limitation of personnel exposure.
- Verify the adequacy of the Equipment Strategy with the vendor's recommendations.
- Ensure that executive action is taken when degradation in site conditions, operating parameters, or procedures is observed.

Provide Hydrogen risk awareness sessions and training to all personnel exposed to Hydrogen risks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 17 May 2024
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Hot work

NARRATIVE:

The ingress of a hydrocarbon slug from the producing wells caused the levels in the treatment section vessels to rise and flood. As a result, unstabilised condensate entered the glycol regeneration train. Finally, the equipment in the glycol regeneration section was flooded and oil was vented through the vent stack. This activated the flammable gas detectors in the surrounding areas and caused the unit to shut down.

WHAT WENT WRONG:

The need to simultaneously bypass multiple layers of protection due to configuration/calibration problems after the initial planned maintenance of the unit, combined with poor risk management, allowed the undetected overfilling of various vessels until condensate venting occurred through the stack of the glycol regeneration section.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

In the case of safety critical equipment inhibition, it is essential to take a holistic view in order to identify multiple inhibits and assess the overall risk level for decision making.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 08 Jan 2024
COUNTRY: Brazil
FUNCTION: Drilling
CAUSE: Explosion, fire or burns
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Hot work
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the well damping operation by bullheading, the pumping pressure reached the configured limit of the relief valve (hydraulic pop-off) of the mud pump, causing its activation and consequently the failure of the primary well barrier envelope. The opening of this valve was not noticed by the team on board and resulted in the loss of hydraulic pressure on the formation. Furthermore, as monitoring parameters were not fully established, as well as a hidden failure in monitoring the pop-off discharge tank, there was a short delay in closing the valves of the subsea workover assembly and Surface Flow Tree (secondary Well barrier envelope), allowing the migration of gas through the production column and DPR (Drill Pipe Riser), reaching the tank room (100% LEL) through the relief valve and, subsequently, discharge on the main deck of the unit through the air exhaust system of the storage room tanks, covering a significant extension of unclassified area (peak of 64% LEL on the sensor close to the well testing plant).

WHAT WENT WRONG:

- Inadequate operating procedure: There was a failure to monitor parameters related to the fluid during the well damping operation.
- Failure in the management and control of critical operational safety elements: There was a late execution of the well closure.
- Failure in the risk identification and analysis methodology: Exhaust from the tank room directed to an unclassified area close to the well test boiler.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include in the Driller's checklist the verification that the tank sensors are activated.
- Communicate to the teams that during the pumping of fluid, the level of the tank used for pumping must be monitored, that is, tank that must lose the pumped volume, as well as the tank that receives the fluid in case of opening the pop-off, and consequently will gain volume in case of opening it.
- Include audible and visual alarm in the driller cabin in case of pop – off valve opening.
- Include questions in the design and operation checklists related to pumping pressure limits in BH (bullheading) operations in LWO (light workover).
- Establish critical operations for on-site monitoring by inspectors, toolpusher and others involved in hydrocarbon risk situations. (Ex: Home well damping, dry test execution, and others.).
- Establish a list of loss of containment evidence for LWO that result in the need for immediate closure of the well. Ex: 1-general alarm of CH4 of unknown origin; 2-Minimum pressure limit on the head at the beginning/during BH; 3-Unexpected return of hydrocarbons at the Well Testing plant and others.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY

In this Section, high potential events reported in the 2024 safety performance dataset that were PSE related but not classified as PSE are presented.

ASIA / AUSTRALASIA OFFSHORE

DATE: 07 May 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Rich glycol liquid overflowed from reboiler still column then soaked inside the reboiler shell insulator and fire was ignited at glycol reboiler shell insulator.

WHAT WENT WRONG:

- No LALL and PAH on glycol surge tank and no LAH at glycol reboiler.
- Glycol reboiler insulation jacket has hole and incompletely seal.
- No routine check valve / vent line blockage verification.
- No specific procedure to alarm on high level at glycol reboiler.
- Lack of competence assurance and supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Replace carbon filter / wet filter at ERPC with appropriate duration.
2. Set up preventative maintenance program to verify glycol reboiler 1” vent line and check valve.
3. Install LAH on glycol reboiler and PAH / LALL on glycol surge tank.
4. Provide training for glycol skid operation and update operating procedure.
5. Conduct process safety upset drill on glycol overspill case.
6. Develop job safety analysis for glycol refilling task to identify risk mitigation before starting job .
7. Revisit glycol reboiler insulation inspection and replacement.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 22 Sep 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Production operator observed light smoke without flame coming out from gas compressor vent fan stack.

WHAT WENT WRONG:

The lube oil was leaking from the lube oil tube and vaporized when touching the hot surface inside the enclosure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Root causes are lack of proper leak test procedure and the re-used tube. Therefore, the investigation team recommends to change the lube oil supply tube of all applicable turbine model after its disassembly and develop the leak test procedure.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 07 Jun 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

During an electrical & instrumentation job at top deck, gas suddenly leaked from a grease nipple of a 6" VB recycle of suction gas compressor. They then informed Supervisor/central control room to decrease speed of gas compressor and control shutdown. Then isolate and bled pressure of valves. At 09:45 hrs. completed bleeding pressure and no continue gas leak from grease nipple.

WHAT WENT WRONG:

Grease nipple of m6"VB recycle of suction gas compressor was cracked.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Implement relief valve and grease nipples survey for all CPP, Phase-1, WHP manual valves. Replacement campaign in December 2024.
2. Check condition of internal check valve and replace per condition in relief valve and grease nipples replacement campaign in December 2024. Use sealant injection new model with cap.
3. Launch CPP manual valve greasing program (planned Q1 2025). Establish PM plan by revisiting manual valve criticality and implement risk-based maintenance.
4. Phase I Unmanned (FEED Study). To expand F&G coverage for unmanned operation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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International Association of Oil & Gas Producers

The Process Safety Event data presented in this report are based on voluntary submissions from participating IOGP Member Companies and are not necessarily representative of the entire upstream oil and gas industry.

The Process Safety Events (PSE) data presented are based on the numbers of Tier 1 and Tier 2 PSE reported by participating IOGP Member companies, and are categorised by:

- onshore and offshore
- drilling and production
- activities
- consequences
- material released

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